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Exploring Generative Artificial Intelligence Perceptions and Information Retrieval
Behaviors Among Students: An Investigation of
Google Search and ChatGPT Queries

Annie Kiele Brown

A thesis submitted to the faculty of
Brigham Young University
in partial fulfillment of the requirements for the degree of
Master of Science

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ABSTRACT

Exploring Generative Artificial Intelligence Perceptions and Information Retrieval Behaviors Among Students: An Investigation of Google Search and ChatGPT Queries

Annie Kiele Brown

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Master of Science

Generative AI tools are rapidly reshaping how undergraduate students seek and engage with academic information, yet empirical understanding of what drives adoption and how students behave when using these tools relative to traditional search engines remains limited. This study examined undergraduate students' information-retrieval behaviors and perceptions when using ChatGPT compared to Google Search, using the technology acceptance model (TAM; Davis, 1989) extended to include trust (Gefen et al., 2003) as the guiding framework. An embedded mixed-methods design was employed, in which Likert-scale perception ratings provided a structured quantitative signal and thematic analysis of semi-structured interviews, structured information-retrieval tasks, and concurrent think-aloud protocols served as the dominant interpretive strand. Ten undergraduate students at a large private university in the western United States completed two counterbalanced sessions, one with each tool. Findings suggest that trust functions less as a coequal TAM construct and more as a gating variable that conditions whether perceived ease of use and perceived usefulness translate into realized advantage. Source transparency emerged as the primary mechanism through which students regulated trust in ChatGPT. Its absence eroded confidence, while its presence restored it. Students calibrated trust thresholds to task stakes and navigated the two tools through structurally different trust-resolution pathways: a layered hierarchy of behaviors on Google and a single-prompting lever on ChatGPT. Notably, ChatGPT adoption continued despite an incomplete trust foundation, a pattern interpreted through Sunstein's (2026) "product trap" framework, in which sustained use is driven less by individual confidence than by external dependencies and competitive pressure. Implications for AI literacy instruction and institutional response are discussed.

Keywords: technology acceptance model, generative artificial intelligence, information retrieval, trust in ai, large language models, technology adoption, information landscape, prompt engineering, AI literacy

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DESCRIPTION OF THESIS STRUCTURE AND CONTENT

This thesis is written in a hybrid format. The preliminary pages of the thesis reflect requirements for submission to the university. The annotated bibliography and IRB approval are found within the appendices.

Introduction

The manner in which students seek and interact with information has undergone a significant transformation with the widespread adoption of the internet. With the advent of the internet, the shift away from traditional library-based to internet-based research is well documented (Technology from Sage, 2023). This trend shows that students today predominantly rely on digital resources rather than physical library borrowing (Coetzee & Stevens, 2025). Students primarily search for information using internet-based tools. However, accessing information on the internet requires a search engine, making these tools an essential gateway to academic information retrieval (IR). Traditional search engines operate through a crawling, indexing, and an algorithmic ranking process that rewards precise keyword construction and source evaluation skills. Searching the internet with keywords has become a basic media literacy skill for students, a skill that has led them to develop specific search behaviors and dispositions (Civilcharan et al., 2015). In fact, developing effective internet search skills has become a fundamental component of academic literacy (Dinet et al., 2012). Without the ability to search strategically and purposefully, students risk retrieving incomplete, irrelevant, or unreliable information. However, how students search for and interact with information online is undergoing another major shift with the introduction of AI.

The advent of large language models (LLMs) powered by generative AI represents a qualitatively different paradigm for accessing internet information. This shift is transforming not only how students search for information but also how results are presented to them. Rather than returning a ranked list of links for students to evaluate and navigate independently, generative AI search systems often present information as conversation responses or synthesized summaries of retrieved information. This changes not only how users search but also how results will be

interpreted and used. The manner in which AI searches the internet collapses the distinction between searching and reviewing, shifting the emphasis from keyword-based search literacy and source evaluation to a process of prompt generation and information synthesis (Lund & Wang, 2023). This marks a fundamental departure from the information retrieval processes students and educators have historically navigated, where the mechanics of the search involved keyword choice, source selection, and credibility assessment were explicit and teachable.

Student adoption of AI tools and AI-mediated information searches has grown rapidly across educational contexts worldwide. Among U.S. teenagers, the proportion using ChatGPT specifically for schoolwork doubled from 13% in 2023 to 26% in 2024 (Sidoti et al., 2025), while a global survey of higher education students across 16 countries found that 86% already use AI in their studies, with over half doing so weekly (Digital Education Council, 2024). In Latin America, 92% of higher education students report using AI, with 67% doing so daily or weekly for school (Digital Education Council, 2024), and broader cross-national data indicating that two-thirds of the general public now use AI tools with some regularity (Gillespie et al., 2025). These patterns suggest that AI-mediated searching and information support are becoming normalized in educational settings. Yet high adoption rates have not translated into confidence or competency, 58% of students globally report feeling they lack sufficient AI knowledge and skills to use these tools effectively (Digital Education Council, 2024). The rapid pace of AI adoption and the change in how AI-enabled internet searches work underscore the need to understand why and how students are using these tools.

Statement of the Problem

Despite the rapid pace of student AI adoption, significant gaps persist in our understanding of what perceptions drive AI adoption and how students behave when using

generative AI for academic information retrieval, relative to traditional search engines. Much of the existing scholarship on Gen AI in education has focused on concerns about academic integrity, misinformation, and the potential over-reliance on AI-generated content (Bushweller, 2024; Lambert & Stevens, 2023; Tindle et al., 2023). While these concerns are valid, they have oriented institutional responses toward restriction rather than understanding, ultimately leaving a meaningful empirical gap in why and how students engage with these tools in practice. Faculty and institutions have largely responded to Gen AI with apprehension rooted in concerns about academic integrity, yet students are already incorporating these tools into their academic workflows, with or without institutional guidance (Dwivedi et al., 2023; Yusuf et al., 2024). This misalignment between student practice and institutional response raises important questions not only about effective AI integration in education but also about students' long-term preparation for the labor market, in which AI competency is an increasingly explicit professional expectation.

Understanding what drives student AI adoption requires a lens capable of capturing subjective experience and perceptual nuance. The technology acceptance model (TAM) proposes that two primary perceptions determine whether individuals adopt and use a technology source: perceived usefulness (i.e., the degree to which a person believes a technology will improve their performance) and perceived ease of use (i.e., the degree to which a person believes the technology will require minimal effort; Davis, 1989). Together, these perceptions shape attitudes toward a technology, behavioral intention to use it, and ultimately, effective use. Subsequent research has extended TAM to include trust as a critical factor, particularly in contexts where users can rely on system-generated outputs they cannot independently verify (Gefen et al., 2003). However, how these factors (i.e., perceived usefulness, ease of use, and trust) operate

specifically in the context of students' academic information retrieval with Gen AI tools remains insufficiently examined.

Purpose Statement

This study examined undergraduate students' information-retrieval behaviors and perceptions when using generative AI tools (i.e., ChatGPT) compared to traditional keyword-based search engines (e.g., Google). Using Likert-scale perception ratings paired with semi-structured interview questions, information retrieval tasks, and concurrent think-aloud protocols with 10 undergraduate students, this study sought to characterize the behavioral and perceptual dimensions of students' Gen AI use in academic contexts and to identify the factors driving adoption of these tools. By grounding findings in both empirical data and the existing literature on information-retrieval and technology adoption, this study contributes to a more nuanced understanding of how students are already engaging with AI and what that reality demands of the institutions responsible for their academic and professional development.

Review of Literature

Information Seeking and Retrieval

Information seeking (IS) and information retrieval (IR) are related but distinct processes that have been studied extensively in academic contexts. While IR refers to the direct, efficient retrieval of specified information through algorithmic systems, IS describes a broader, often exploratory process through which individuals actively construct understanding across knowledge gaps (Fadaie, 2012). Foundational models proposed by Wilson (1981, 1999), Dervin (1983, 1992), Ellis (1989), Ellis et al. (1993), and Kuhlthau (1991) suggested that IS is not a simple transactional exchange but a complex, iterative process shaped by individual cognition, context, and the affordances of available tools. In academic settings, students' IS behaviors have

historically been shaped by the dominant IR tool of their era (e.g., print-based classification systems and card catalogs to digital databases and web-based search engines). Research has documented that students gravitate toward tools they perceive as familiar and easy to use, often prioritizing accessibility and speed over depth or scholarly rigor (Civilcharran et al., 2015; Dinet et al., 2012). The arrival of LLM-powered generative AI tools has introduced a qualitatively different IR paradigm, one that demands re-examination of these well-established behavioral patterns.

Search Engines & Generative AI

Unlike traditional search engines, generative AI search tools do not return ranked links to existing sources; they synthesize and generate novel responses in direct, conversational dialogue with the user (Lund & Wang, 2023). Schuetzler et al. (2024) found that students interacting with generative AI engaged in iterative “prompt cycling,” progressively refining queries through conversational exchange in ways that differ fundamentally from discrete keyword-based queries. Gibbons et al. (2023) similarly documented a behavioral tendency toward efficiency-driven summary-retrieval in AI interactions, with users iteratively prompting AI to compress its own outputs; a pattern reflecting a prioritization of cognitive ease and speed that is consistent with, yet behaviorally distinct from, student behaviors documented in traditional search contexts. While these studies illuminate emerging patterns in students’ Gen AI use, a mixed-method empirical investigation of how these behaviors compare directly with traditional search patterns remains limited.

Technology Adoption

Understanding what drives students’ adoption of generative AI for academic information retrieval requires a theoretical framework that accounts for the perceptual and attitudinal factors

underlying technology use. The TAM introduced by Davis (1989) proposes that two primary perceptions determine technology adoption: perceived usefulness (PU), the degree to which a user believes a technology will enhance their performance, and perceived ease of use (PEOU), the degree to which they believe it will require minimal effort. TAM has demonstrated robust explanatory power across diverse educational technology contexts (Granić & Marangunić, 2019), and its core constructs map directly onto the efficiency and effectiveness factors students report when describing their use of Gen AI. Gefen et al. (2003) extended TAM to include trust as a third critical variable, arguing that in contexts where users rely on system-generated outputs that they cannot independently and easily verify, trust significantly shapes both adoption and continued use. This extension is particularly salient for generative AI, where students interact with synthesized outputs whose sourcing and accuracy are not transparently verifiable.

Trust & Perception

Research on student trust in generative AI has yielded nuanced findings. Baek and Kim (2023) found that users were more likely to trust ChatGPT when it demonstrated task efficiency and personalized its responses. This suggests that trust is actively cultivated through satisfying user experiences rather than being a fixed prior disposition. Mayerhofer et al. (2025) observed that users frequently accepted AI-generated responses simply because they appeared well-formatted and plausible, even without verifying accuracy; a pattern with significant implications in academic contexts where hallucinated citations and fabricated information had been documented risks (Bender et al., 2021). C. Baek et al. (2024) found that college students often described ChatGPT outputs as “too good to be true,” yet some students continued to use the tool, reflecting a complex, inconsistent relationship between skepticism, trust, and tool use in AI-generated information. Sunstein (2026) offers a useful theoretical lens for interpreting this sort of

pattern, introducing the concept of “product traps” in the World Happiness Report 2026. Product traps describe situations in which individuals continue using products that they believe may reduce their own well-being, often because their use is shaped by external dependencies rather than purely individual preference or perception. In the context of social media, Sunstein shows that many users persist in using platforms whose abolition they would pay for, highlighting a divergence between behavior and perceived welfare.

Applied to generative AI in academic contexts, this framework does not suggest that AI use is welfare-reducing, but it does raise an important analytical possibility. That students’ continued adoption and use of Gen AI tools may not fully reflect trust or satisfaction, but may instead be sustained by perceived efficiency advantages or shifting academic norms. Despite growing evidence of these perceptual dynamics, mixed-method research examining how trust, perceived usefulness, and ease of use jointly shape undergraduate students’ adoption of and engagement with generative AI for academic information retrieval remains insufficient.

Research Questions

This study addressed the following research questions:

1. How do undergraduate students’ perceived ease of use, perceived usefulness, and perceived trust in generative AI tools influence their academic information retrieval behaviors and adoption decisions?
2. What information retrieval behaviors do undergraduate students employ when using generative AI tools, and how do these behaviors differ from those employed when using traditional keyword-based search engines?

Methods

This study employed an embedded mixed-methods design in which qualitative methods

were prioritized, and a quantitative descriptive strand was used to provide a supplementary perceptual signal (Creswell, 2008; Kimmons, n.d.). This design is appropriate when one method is dominant, and a second method is used to support or contextualize it rather than to answer an independent research question of equal weight. Here, structured Likert-scale perception ratings served the embedded quantitative role, generating item-level descriptive comparisons across tools that anchored and oriented the qualitative analysis. The dominant strand (thematic analysis of participants' verbal explanations and think-aloud session recordings) provided the primary interpretive evidence for answering the study's research questions.

This design is well-suited to the study's aims. Understanding what perceptions drive student adoption of generative AI tools requires a framework capable of capturing subjective experience, attitudinal nuance, and behavioral variability, dimensions that qualitative inquiry is uniquely positioned to illuminate (Merriam & Tisdell, 2016). At the same time, the inclusion of structured Likert items grounded in the technology acceptance model (Davis, 1989; Gefen et al., 2003) provided a theoretically mapped, item-level structure against which qualitative themes could be interpreted. The two strands are therefore integrated by design, with the quantitative ratings establishing a descriptive baseline organized around TAM constructs, and the qualitative analysis explains and expands upon what those ratings and behavioral patterns mean.

Participants and Sampling

Ten undergraduate students at a large private university in the western United States participated in this study. Participants were recruited using purposeful sampling (Marshall, 1996) to identify individuals with prior experience using both traditional internet search engines and generative AI chatbot tools. This approach prioritizes depth of insight over sample breadth, consistent with the goals of mixed-method inquiry, where the qualitative strand is emphasized, in

which a small number of key informants yields in-depth descriptive data, prioritizing data saturation over statistical significance (Hennink & Kaiser, 2022). Of the 10 participants, six were female, and four were male. Participants represented three class levels: five freshmen, two sophomores, and three seniors. Participants self-reported their expertise with each tool on a four-point scale (*Novice, Beginner, Intermediate, Expert*). Reported expertise differed importantly between the two tools. For Google, participants skewed toward higher expertise, with five identifying as Expert, three as Intermediate, and two as Beginner. For ChatGPT, participants skewed toward lower expertise, with seven identifying as Beginner, one as Novice, and two as Intermediate, with no participant identifying as Expert. All participants provided informed consent prior to participation, and confidentiality was maintained throughout the study. Each of the participants is identified using a participant ID as identified in Table 1.

Table 1

Participant Characteristics

Participant ID	Gender	Class Level	Google Expertise	ChatGPT Expertise
P1	Male	Freshman	Beginner	Intermediate
P2	Female	Freshman	Beginner	Novice
P3	Female	Freshman	Intermediate	Beginner
P4	Female	Senior	Intermediate	Beginner
P5	Female	Sophomore	Expert	Beginner
P6	Male	Freshman	Expert	Beginner
P7	Male	Freshman	Intermediate	Intermediate
P8	Female	Senior	Expert	Beginner
P9	Male	Sophomore	Expert	Beginner
P10	Female	Senior	Expert	Beginner

Data Collection Procedures

Data were collected using three integrated methods: semi-structured interviews paired

with Likert-scale perception ratings, followed by structured information-retrieval tasks, and concurrent think-aloud protocols. All sessions were conducted remotely via Zoom video calls, with participants using their own devices and accounts to access the designated tools. Zoom recordings served as the primary means of capturing participant verbalizations, screen activity, and research-participant exchanges. ChatGPT conversation logs were additionally collected to provide a documentary record of each participant's interaction patterns and prompt strategies.

To reduce the potential for task-order bias and participant fatigue, data collection was divided into two sessions, with one for each tool (Google and ChatGPT), each lasting approximately 30-45 minutes. Sessions were scheduled based on participant availability, typically within a few days of each other. The order in which participants completed the Google and ChatGPT sessions was counterbalanced across the sample, so that approximately half of the participants completed the Google session first and the other half completed the ChatGPT session first. This design mitigates the confounding influence that prior experience with one tool may affect perceptions of the other.

Likert Scale Perception Ratings

Responses were recorded on a 6-point scale ranging from 1 (*Strongly Disagree*) to 6 (*Strongly Agree*), with intermediate points of *Disagree* (2), *Somewhat Disagree* (3), *Somewhat Agree* (4), and *Agree* (5). A 6-point scale was used in lieu of a 5-point scale to eliminate a neutral midpoint, thereby requiring participants to indicate a directional position on each item (Garland, 1991). Items were designed to measure perceptions of tool ease of use, convenience, speed, usefulness, accuracy, and trustworthiness. These are constructs that map directly onto the TAM dimensions of Perceived Ease of Use (PEOU), Perceived Usefulness (PU), and Trust (Davis, 1989; Gefen et al., 2003). Each item was aligned with one of these three constructs based on

alignment with Davis's (1989) definitions and Gefen et al.'s (2003) trust extension.

The items "easy to use" and "convenient to use" operationalize PEOU, reflecting Davis's definition of PEOU as the degree to which a user believes a system will require little effort. The items "finds information quickly" and "provides useful information" operationalize PU, consistent with Davis's framing of PU as the degree to which technology enhances task performance. The items "provides accurate information" and "provides trustworthy information" operationalize Trust and correspond to Gefen et al.'s competence and integrity dimensions, respectively. Follow-up probes were used where responses warranted further elaboration.

Structured Information-Retrieval Tasks

During each session, participants completed a series of structured information-retrieval tasks using the assigned tool for that session. Tasks were designed to mirror the types of academic information-retrieval activities students commonly encounter, spanning basic recall, concept explanation, training and instruction, resource identification, and fact-checking.

The five task types were held constant across both sessions, with the specific subject matter changed between tools to prevent practice effects. The basic recall task asked participants to identify a foundational text by a specific author and summarize its central ideas (Patrick Süskind's first novel for Google; Descartes's first philosophical treatise for ChatGPT). The concept explanation task asked participants to explain a specialized academic concept (the Gettier Problem; Negative Capability). The training and instruction task asked participants to identify a named procedure and describe how it is performed (the Gaskin Maneuver; the Epley Maneuver). The resource identification task asked participants to identify a non-mainstream organism and provide five sources about it (tardigrades; blanket octopuses). The fact-checking task asked participants to evaluate three claims of varying historical or scientific plausibility,

with both sets including one item about the Voynich Manuscript that served as an internal anchor across sessions.

Participants verbalized their responses and thought processes aloud as they worked through each task, and no separate written submission was required. Sessions were moderated by the researcher, who observed and occasionally asked probing questions to elicit further insight into participants' information retrieval behavior. Upon completing all tasks within a session, participants shared a general reflection on their overall experience with the tool, allowing for the capture of both immediate, task-specific perceptions and a holistic impression of each tool.

Think-Aloud Protocol

Participants were asked to verbalize their thought processes throughout task completion using a concurrent think-aloud procedure, providing real-time access to how they evaluated the usefulness, ease of use, and trustworthiness of each tool's outputs. This method captured spontaneous articulations of perceived tool value that might not surface through post-hoc self-report alone (Ericsson & Simon, 1993). Zoom recordings preserved these verbalizations for subsequent analysis.

Data Analysis

Data analysis proceeded in three sequential phases, each building on the preceding layer of evidence. In the first phase, Likert-scale rating data were analyzed descriptively to produce an initial quantitative signal of participants' perceptions across the three TAM constructs: Perceived Ease of Use (PEOU), Perceived Usefulness (PU), and Trust (Davis, 1989; Gefen et al., 2003). Ratings were examined item by item and organized into paired comparisons across tools, providing a structured baseline for interpreting patterns in subsequent qualitative data.

In the second phase, participants' verbal explanations of their Likert ratings were

subjected to thematic analysis (Braun & Clarke, 2006). As each rating was accompanied by a spontaneous verbal justification, this layer of analysis allowed for construct-level interpretation grounded in participants' own accounts of their perceptions. Themes were identified within each item and then examined across all participants and items, enabling cross-participant comparison anchored to a common referent.

In the third phase, full session recordings were analyzed to examine participants' broader information retrieval behaviors and perceptual patterns beyond the rating items. Research memos and observational notes recorded concurrently during data collection served as a basis for initial codes, consistent with Saldaña's (2021) recommendation that early analytic impressions be systematically documented to preserve interpretive proximity to the data. These preliminary codes were applied to the full data corpus, and open coding permitted additional categories to emerge inductively from the data. Codes were then examined across all participants using the constant comparative method (Glaser & Strauss, 1967), through which recurring patterns were refined into overarching observational themes that cut across individual cases. Member-checking and peer debriefing were employed throughout to strengthen the credibility of the analysis (Lincoln & Guba, 1985).

Results and Discussion

Initial Perception Responses

To determine students' views regarding ease of use (PEOU), Perceived Usefulness (PU), and Trust, participants were asked to give an initial rating of their perceptions (see Tables 2 and 3) and then explain their position.

Table 2*Likert Responses Summary by Search Resource*

Questions	Strongly Agree	Agree	Somewhat Agree	Somewhat Disagree
Google is easy to use	3	7		
ChatGPT is easy to use	3	5	2	
Google is convenient to use	8	2		
ChatGPT is convenient to use	2	6	2	
Google: Find information quickly		5	5	
ChatGPT: Find information quickly	2	5	3	
Google: Provides accurate information	2	6	2	
ChatGPT: Provides accurate information		5	5	
Google: Provides useful information	2	8		
ChatGPT: Provides useful information	2	5	3	
Google: Provides trustworthy information	1	7	2	
ChatGPT: Provides trustworthy information		1	8	1

Note. None of the respondents selected Disagree or Strongly Disagree

Table 3*Construct (PEOU, PU, Trust) Paired Comparisons*

Item	Google	ChatGPT	Tie	Preferred Tool
Easy to use	4	2	4	Google
Convenient to use	6	0	4	Google
Find information quickly	1	4	5	ChatGPT
Provides useful information	3	0	7	Google
Provides accurate information	6	2	2	Google
Provides trustworthy information	7	0	3	Google

Respondents rated Google higher than ChatGPT on the PEOU construct items (easy to use and convenient to use), though the strength of that preference varied by item. In terms of ease-of-use, respondents' perceptions of both tools were generally positive. All ten participants

agreed or strongly agreed that Google was easy to use. Eight of the ten participants agreed or strongly agreed that ChatGPT was easy to use. The remaining two respondents “somewhat agreed” that ChatGPT was easy to use. A paired comparison revealed that four participants rated Google higher, two rated ChatGPT higher, and four rated both tools equally easy to use, suggesting a modest preference for Google on this criterion (see Table 3)

In terms of the convenience item, the response pattern differences were more pronounced. Eight participants strongly agreed, and two agreed that Google was convenient to use. For ChatGPT, two participants strongly agreed, and six participants agreed that ChatGPT’s search was convenient. Paired comparisons of these data showed that six participants assigned Google a more favorable rating for convenience. None of the respondents felt ChatGPT was more convenient. Four respondents felt the two tools were equally convenient (see Table 3).

For PEOU overall, these patterns demonstrate a slight preference for Google. However, participants’ explanations of their ratings reveal that the ease and convenience preferences associated with Google and ChatGPT were grounded in how well the tool had been integrated into the technology they use and how familiar they are with interacting with each tool. Three themes emerged that illustrate this distinction and its implications for understanding PEOU in the context of AI-mediated academic information retrieval.

For perceived usefulness (PU), participants’ preferences followed a pattern distinct from that of the PEOU items. On the item addressing retrieval speed, ChatGPT emerged as the preferred tool, being the only item across the full dataset where that is the case. Four participants rated ChatGPT higher on speed, one rated Google higher, and five rated both tools equally. ChatGPT’s ratings on this item tended to be more positive, with two participants selecting Strongly Agree and the remaining ratings distributed across Agree and Somewhat Agree.

Google's ratings on the same item, by contrast, were distributed evenly between Agree and Somewhat Agree, with no participant selecting Strongly Agree (see Table 2).

The useful information item produced a markedly different pattern. Google's ratings were uniformly positive, with all ten participants selecting either Agree or Strongly Agree. ChatGPT's ratings on the same item included three participants at Somewhat Agree, reflecting a modest but consistent tendency toward qualified rather than unequivocal endorsement. A paired-comparison of this item revealed that three participants rated Google higher in usefulness, no participant rated ChatGPT higher, and seven rated both tools equally (see Table 3).

Across both Trust items, the divergence between Google and ChatGPT was more pronounced than the other constructs. Google's ratings on the accuracy item were distributed across Strongly Agree, Agree, and Somewhat Agree, with eight of ten participants selecting Agree or Strongly Agree. ChatGPT's ratings on the same item split evenly between Agree and Somewhat Agree, with no participant selecting Strongly Agree. The trustworthiness item produced the most striking pattern in the entire dataset. Google's ratings concentrated at the positive end of the scale, with eight of ten participants selecting Agree or Strongly Agree and only two selecting Somewhat Agree. ChatGPT's ratings were dominated by Somewhat Agree, with 8 of 10 responses, while only 1 participant selected Agree and 1 selected Somewhat Disagree (see Table 2). Paired-comparison results confirmed this as the most one-sided across all items, with seven participants assigning Google a more favorable trustworthiness rating than ChatGPT, and no participant doing so for ChatGPT (see Table 3).

In summary, there seems to be a slight preference for Google when retrieving information. With the exception of information retrieval speed, participants' responses suggest that Google searches might be slightly easier to use, more convenient, and that they trust results

slightly more than those from ChatGPT.

Theme Extraction

While initial survey response patterns suggest a slight preference for Google when retrieving information, respondents' explanations of their initial perceptions provide a clearer picture of their perceptions of these information retrieval tools. Based on an analysis of interview responses, specific themes emerged that clarify participants' perceptions. Some of the themes extracted from this analysis are based on a specific construct; others, however, overlap or integrate various aspects of ease, convenience, and trust.

Theme 1: Tool Integration. Participants described Google search as immediate, effortless, and deeply integrated into everyday technology. In this sense, ease of use was often evaluated as easy to access. Participants emphasized that Google is readily available across devices and platforms. Respondent P9 observed that “you always have your phone and Google at your fingertips,” and respondent P6 noted that Google is accessible “basically anywhere,” including through tools like Siri and Alexa. Respondent P10 reinforced this point by saying, “I use it almost every day for a variety of different things; it’s always there.” In these accounts, convenience was closely linked to tool integration. Google was perceived as easy to use because it was already embedded in the technologies participants used daily. ChatGPT’s ease of use, by contrast, was more often described as still under development rather than as an immediately available part of the technology they used.

Ease and convenience, as measured by accessibility and integration, were central to respondents' perceptions of these tools. Google has been widely used for decades, and educational institutions have often incorporated information retrieval and media literacy into academic practice as an organic response to a tool that had already become ubiquitous in

students' daily lives (Sundin et al., 2017). In addition, Google's continued investment in interface simplicity has kept the barrier to entry low across successive iterations of the platform, reinforcing ease perceptions through design and accessibility. Research has long established that developing effective internet search skills is a fundamental component of academic literacy (Dinet et al., 2012) and that students gravitate toward tools they perceive as familiar and accessible (Civilcharran et al., 2015). The ease with which participants reported using Google reflects the convergence of these factors. First, an interface designed for low friction, reinforced by an educational environment that has come to treat search proficiency as a baseline competency. It should be noted that at the time of this research, AI search capabilities had not yet been integrated in the same manner as they are now.

Theme 2: Search Literacy and Prompt Engineering. A second theme that emerged with notable consistency was that PEOU was not only related to the tool's accessibility but also to how easy it was for users to produce results. In this regard, a user's PEOU was often associated with their familiarity and experience with the tool. The respondents in this study reported their skill in conducting Google searches as intermediate or expert. In contrast, eight of the 10 respondents described their ChatGPT experience as novice or beginner. This influenced their perception of ease and convenience.

As respondent P4 explained, "[Google is] really user-friendly ... you only have to use the search bar," and type in a few key phrases to get a list of possible answers. Respondent P1 noted that once search terms are entered, Google "kind of does the rest for you." Other participants used similarly direct language, describing Google as "simple" (P6) and "not difficult to use at all" (P2). These responses suggest that Google's ease of use is tied not only to interface simplicity but also to how easy it is to get results. However, some respondents also noted that

Google's ease of use depended on participants' expertise and experience with searching and selecting results.

Several participants explicitly framed ease of use as knowing how to initiate an effective search to get what they needed. They tended to get quick results with a simple keyword search. However, respondent P1 noted that "the more specific your question is, the more likely you're able to find what you're actually looking for," positioning query precision as the governing condition for retrieval success. Respondent P9 offered a concise statement of this conditionality: "there's lots to sort through, and if you know how to use a lot of different search options, then yeah, it's easy." For these respondents, ease of use depends on the user's command of Google's search techniques and the use of appropriate keywords. Respondent P2 described the consequence of this skill's absence, noting that,

sometimes I'll have a very specific question, and I'll look it up, and [Google] won't give me the exact answer I'm looking for. Then I kind of have to click on a few websites and read stuff and figure it out myself,

suggesting that Google will easily give results regardless of how well you ask the question, but you get better results if you know how and what to ask. If you give a poor search request, you will get results; you just need to look through the results listing more carefully. Respondent P4, whose professional role explicitly involves fact-checking, framed this as a practiced and accumulated skill: "you have to kind of tweak the search to figure out what you want to get out of the search results, and I've just had experience with that," describing ease not as a function of accessibility or the tools' interface design alone, but in the expertise and knowledge of searching the user brings to it.

In contrast, ChatGPT's ease of use depended on developing the new skill of prompt

engineering. Respondent P8 articulated this requirement most directly and with the greatest specificity: “you have to know which specific prompt to give it. And if you don’t understand how to do that, you just get whatever it answers,” adding that through experience, they had learned to “give it a role and a target ... something it needs to accomplish and the specific actions it needs to do,” framing effective prompting as a learnable but non-obvious technical skill. The effect on one’s PEOU is captured in respondent P8’s observation that, “prompting takes a long time. Half of ChatGPT is coming up with the right prompt in order to get a good response, and that takes time. With Google, you can put in whatever you want.” This comparison is telling: Google tolerates imprecise input because the index absorbs it and still returns something navigable, while ChatGPT requires deliberate input construction because the output depends on the instructions you give the AI. Respondent P6, despite selecting “Strongly Agree” on the ease of use question for ChatGPT, acknowledged the same condition: “the ease of use mostly comes from using the right prompts...I find that giving it a lot of context helps it create the answer that you’re looking for,” indicating that even highly favorable users attributed ease partly to developed prompting skill. Respondent P3 framed this as an evolving competency that needs to be developed: “it’s kind of hard to get your point across and what you’re asking sometimes, but I also feel like maybe that’s a skill that you can develop over time. I know some people who can get answers right away, exactly how they want them, but I haven’t ever been able to do that yet,” situating the learning curve within a developmental trajectory rather than treating it as a fixed barrier. Respondent P9 extended this to a recognition of this limitation as a dimension of prompt literacy, “I could have prompted it differently. There is a learning curve.” These responses reveal a more sophisticated layer involving skill development as a function of ease of use, in which effective use requires not only knowing how to prompt but knowing what

to prompt for.

The parallel structure of these findings carries meaningful implications for how PEOU is understood in AI-mediated information retrieval contexts. Both tools place a skill burden on the user, but at different points and of different types. Google's demand is concentrated mostly in source evaluation, which operates on the logic of filtering and selection. ChatGPT's demand is embedded in prompt design and an awareness of the tool's functional boundaries, which operate on the logic of instruction generation. The fact that participants attribute ease of use to both tool design and accessibility, while also acknowledging that effective use depends on skill, suggests that students may already be treating prompt engineering the way they treat search literacy, as a normal part of getting what they need from a tool.

A meaningfully distinct dynamic emerged with prompt engineering replacing search literacy as the primary skill required to make the tool easy to use. The difference is not simply one of label but of underlying logic. Google's search literacy operates on a retrieval model: the user crafts a keyword query to locate pre-existing content, and success means finding the right document within an index that already contains it. Prompt engineering, by contrast, operates on a generative model: the user creates instructions that shape what is produced. Success is dependent on eliciting a synthesized response that did not exist before the interaction. This distinction matters because it reframes what "getting it wrong" means in each context. A poor Google search returns irrelevant links. Respondent P1 acknowledges that, without specificity, Google's breadth becomes a liability, leading the platform to return large volumes of irrelevant responses that need to be reviewed. A poor ChatGPT prompt produces a response that may appear correct and complete while failing to address what the user actually needs, a failure that is harder to detect and correct without understanding why it occurred. Ease of use in a Google search is a

function of knowing how to craft the keyword query to get related results. Ease of use in AI search means having experience with engineering prompts. Most participants reported a lack of experience creating AI prompts and, consequently, found it more difficult to conduct an AI search than a Google search.

Notably, however, participants did not describe their interactions with ChatGPT using the language of prompt engineering. Rather, they articulated developing the prompting skill through trial and error, describing processes of adjusting, refining, and reattempting until a satisfactory response emerged. Respondent P3 described this, stating, “I think it’s an awesome idea that it can help us with so many things. But I feel like there is a learning curve to getting what you need out of it,” framing the process as exploratory rather than strategic. Respondent P8 similarly described the early experience of using ChatGPT as requiring active exploration before any fluency develops. He said, “you have to kind of go out of your way to use it ... you have to explore ChatGPT first to understand it,” suggesting that the path toward ease is itself an unguided discovery process. None of these participants identified what they were doing as prompt engineering, a recognized and teachable skill with its own vocabulary and strategies, underscoring a broader institutional gap: students are developing prompt engineering competency organically, without the scaffolding that would help them recognize it as a transferable, nameable skill. Given that prompt engineering and keyword search literacy are not simply different degrees of the same ability, but distinct cognitive practices suited to different retrieval architectures, this gap represents precisely the space where targeted AI literacy instruction could have its greatest impact.

Theme 3: Purpose Appropriateness. A third theme related to PEOU was articulated as purpose appropriateness (e.g., what they were trying to accomplish). With regards to PEOU,

participants consistently drew explicit distinctions between the types of tasks each information retrieval tool was well-suited to accomplish. Using the correct tool for the task was important to perceived ease of use. For example, Respondent P1 framed this in terms of task seriousness: “[ChatGPT is] convenient for schooling and business things, more professional things. I would use Google if it were something small and stupid. ChatGPT can help me with that serious stuff, whereas Google probably wouldn’t be able to even understand,” positioning ChatGPT as the preferred, easier-to-use, tool for substantive, complex information retrieval tasks while reserving Google for quick, low-stakes queries. Respondent P4 drew the sharpest task-based distinction, separating factual retrieval for academic research as categorically different use cases, noting that,

small things like definitions, geographical stuff, or just simple no-brainer questions are easier and quicker to look up. And then if you want to go into specific topics and dive into those to get a hyper-specific answer, that just takes a little bit longer.

Respondent P9 echoed the conditional framing: “for super basic stuff like just checking in, it doesn’t have to be super rigorous, then yeah, [Google is] pretty convenient,” simultaneously acknowledging that this convenience did not extend to tasks requiring rigor or in-depth curating of information to answer a specific question. These accounts reveal that ChatGPT’s ease and convenience were high for exploratory, contextually-nuanced tasks that required synthesized answers from various sources. Respondents had a more favorable PEOU when the information retrieval task could be handled well by the resource.

Theme 4: Organic Search Results and Verification Time Requirement. The issue of how and which results were presented also affected PEOU. The issue of organic search results was specific to Google and was expressed as a frustration rather than satisfaction. With Google information searches, the issue of result relevance emerged primarily around the friction

introduced by sponsored content and search engine optimization. Participants identified this as a meaningful irritant. Respondent P4, for example, described this friction in practical terms:

[Google is] going to pull up the stuff that has paid the most to put their product up at the top instead of the thing that is the best for what I'm looking for. ... I have to scroll past all of those to get what I actually want or need,

identifying commercial interference as a direct obstacle to retrieval ease, which required active navigation to overcome. Respondent P6 situated the same concern within the context of deeper research, specifically:

I think sometimes with Google, deeper research and things like that can be hard to find the right sources, and often because of SEO [Search Engine Optimization], I feel like a lot of the wrong kind of websites come to you sooner than the ones you really need, drawing a clear distinction between casual queries, for which Google performs reliably, and academic or research-oriented queries, for which SEO may introduce friction. Respondent P8 extended this to a concern regarding result bias: "sometimes the information I want is not what I [get] because of sponsored stuff. It's usually biased towards someone paying for an advertisement." They framed the problem not merely as an inconvenience but also as a distortion of the result set that undermines the relevance of what surfaces. Respondents felt Google was less convenient because they had to sift through irrelevant search results produced as a result of an SOE algorithm. However, participants still rated Google as easy and more convenient overall, suggesting that commercial interference was experienced as an irritant embedded within an otherwise easy-to-use free platform for retrieving information, rather than as fundamental challenges to its usability. PEOU was affected when commercial forces distorted the output, requiring users to filter results they perceived as potentially biased because they were sponsored.

The equivalent ease-of-use issue for ChatGPT revolved around the time it took users to review and verify results. In ChatGPT, the commercialization of internet searching as a PEOU issue was not mentioned. Rather, participants noted the time required to verify results.

Respondent P5 articulated this most directly across two separate items, saying, “if I’m using ChatGPT for research or finding sources, it adds a step. I could just Google it from the beginning, and it would have been faster,” and “if I’m trying to find valid sources for school, then ChatGPT takes longer than Google. It’s adding an extra step because I end up on Google anyway” to verify the sources which affects convenience. This is a trust issue; however, the belief that an AI search requires a verification step negates any convenience gained from receiving a direct synthesized response. Respondent P2 downplayed this concern based on the fact that AI is still developing: “sometimes I know it doesn’t give the right information, and it can be a hassle... it still has ways to go to get better,” framing reduced convenience as a present but temporary feature of an emerging technology. When this happened, the perceived convenience of receiving a direct response to questions curated from multiple sources was diminished because the cost of verifying a response offsets any convenience gained.

Theme 5: Conversation as a Distinctive Aspect of Usability. Among the qualities participants attributed to ChatGPT, its conversational format emerged as its most genuinely distinctive strength, one that Google’s keyword-based retrieval model was not designed to replicate. While respondents indicated that ChatGPT is perceived as being fast, this theme identifies a dimension of usefulness rooted not in the speed with which results are obtained but in the tool’s capacity to revise, explain, iterate, and engage with the results in a fundamentally different way.

The most basic expression of this difference is that ChatGPT does not merely locate

information; it interprets and contextualizes it. Where Google returns links ranked by potential relevance, ChatGPT adds an explanation to the information retrieval process. As respondent P7 noted, it “not only gives you exactly what you’re looking for, but also explains what you’re looking for,” a distinction that represents a meaningful expansion of what usefulness can mean in an information-retrieval context. This explanatory quality is reinforced by natural language interaction. Rather than requiring users to construct precise search terms, ChatGPT responds to how people naturally describe their needs. Respondent P10 captured this when describing the experience as “like having a conversation, then getting exactly what you’re looking for,” suggesting that conversational input produces more precise and satisfying outcomes than keyword construction typically allows.

This advantage becomes most pronounced in tasks that are nuanced, language-dependent, or difficult to reduce to a search query. Respondent P1 identified academic writing as one such case, observing that when a sentence “just didn’t sit well,” ChatGPT could help rework it in ways Google “probably wouldn’t be able to even understand.” Here, the tool’s usefulness is not incidental but structural. ChatGPT is doing something Google keyword retrieval is architecturally unsuited to do.

A further and related strength is ChatGPT’s capacity for iterative dialogue. Respondent P4 described the value of being able to “talk back and forth with it to figure out your ideas and what you want to do,” positioning usefulness not as something delivered in a single response but as something developed through exchange. This makes ChatGPT particularly well-suited to generative or exploratory tasks such as refining an argument, developing an idea, or improving a piece of writing, where the goal itself may evolve over time.

However, this same depth carries a limitation. When poorly calibrated to the nature of a

request, ChatGPT's explanatory tendency can become counterproductive. Respondent P2 reported receiving "8 paragraphs" of information in response to a simple query, describing it as "too much." This suggests that conversational depth is not inherently valuable, its usefulness depends on how well the tool adjusts the scope of its response to the complexity of the task. For straightforward queries, over-explanation introduces noise rather than clarity.

Taken together, these findings suggest that ChatGPT's distinctive advantage lies in tasks that are language-dependent, iterative, and generative. This does not position ChatGPT as universally superior, but it does identify a qualitatively distinct form of usefulness that keyword-based search cannot currently offer.

Theme 6: Speed, Quality, and Trust. Participants consistently identified ChatGPT as the faster tool, attributing this to its ability to synthesize information into a single response, allowing users to converse with the AI rather than requiring them to navigate reviews and filter multiple sources. Yet speed alone did not determine perceived usefulness. Speed was not treated as a proxy for quality. The two dimensions were weighted separately, and when accuracy was in doubt, the efficiency advantage lost much of its force. As Participant 2 put it, "ChatGPT can give stuff fast, but that doesn't mean I really trust the accuracy." Or as respondent P5 stated, "usefulness depends on accuracy, so if I don't trust it fully, I won't use it as much." This is reinforced by respondent P3, who described the lack of trust as diminishing ChatGPT's speed advantage, saying, "it definitely makes me want to read multiple different sources, which makes it so it's no longer quick." This distinction between a response that arrives quickly and one that is genuinely useful is the central finding across both perceived usefulness items.

In general, how quickly a response was provided was seen as an independent but insufficient dimension of usefulness. ChatGPT quickly and impressively gave a response in

context. However, even if the response seemed reasonable, several participants indicated they didn't trust the result. What this reveals is that speed carries perceived value on its own terms, independent of output quality. Respondent P7 made the comparative advantage explicit, stating that,

I have to spend a longer time to find an answer when on Google, but I believe it will be more credible and useful because it isn't automatically written ... you'll find something from someone who is scholarly and put effort into it.

The slower retrieval process was not simply tolerated; it was interpreted as a meaningful signal of credibility. Respondent P6 articulated this saying,

I like the fact that maybe Google makes me kind of read a little bit more and do my own research. I feel like when it makes me do that, the information sticks better ... it's better for me to retain and remember.

In this framing, the time Google demands is not inefficiency; rather, it is evidence of the intellectual labor participants associate with trustworthy information. This argument extended the logic into the domain of learning, arguing that Google's requirement to read more and conduct independent research made information easier to retain. What might otherwise appear as a weakness of Google was recast as educational productivity. In contrast, ChatGPT was described as producing high volumes of potentially low-quality results that needed verification. For some, ChatGPT-generated content lacked source transparency and human authorship, something students associated with genuinely useful answers.

The implication is that users trusted the list of responses Google provided as credible sources. They did not trust the AI to provide a credible summary of the sources. So, while Google consistently required more time and effort from users to summarize and synthesize

information, it seemed better than having the AI do it for them.

From this, a consistent pattern emerged: students did not treat speed and quality as equivalent or interchangeable markers of usefulness. They weighed them differently. What determined which dimension prevailed was trust. When confidence in a tool's accuracy was strong, speed amplified its usefulness. When that confidence was absent, speed became largely irrelevant. ChatGPT was faster, but Google was seen as more credible. They also saw educational value in the effort required to extract and summarize useful information from the list of potential sources Google provided.

Trust Theme 7: Source Transparency as the Primary Trust Failure Issue. The most consistent finding across all 10 participants was that source transparency was the central factor in how students evaluated trust in ChatGPT. Operating in two directions, the absence of transparency caused trust to fail, and its presence restored it.

When sources were absent, participants identified this as the core obstacle to trust.

Respondent P6 articulated this issue, saying,

I have no idea if it's accurate because I don't know where it's sourcing things from.

Potentially, the AI believes it to be real, or it's taking it from somewhere that presents it as real, but I can't tell where the information is coming from.

Respondent P3 described a lack of trust even when ChatGPT provided sources, "there's always that question of is this really credible? And it feels like you need to do more digging than just letting the computer say that this is correct." Respondent P10 identified source provision as a feature whose absence was actively felt: "the sources are missing; if they were provided, it would give me more peace of mind. It's something I would like to see."

The issue was not simply that ChatGPT might be wrong, but that its single synthesized

response offered no pathway for independent verification. Respondent P4 expressed the opinion that “on the internet, you can personally fact-check it yourself because you can look at where those sources got their information.” Unlike Google, where users can trace and evaluate information themselves, ChatGPT’s confident, polished outputs left students with no clear basis for assessing accuracy. Paradoxically, ChatGPT’s perceived strength, fluent information synthesis, was precisely what undermined informed trust. However, when ChatGPT provided the sources it used, trust increased immediately and consistently. Participants reported feeling more confident, finding ChatGPT more credible, and viewing it as more comparable to Google as a reliable information source. Several noted a meaningful shift in perception after discovering that ChatGPT could provide sources on request. For example, respondent P2 described a formative shift in perception when sources were provided, saying, “I trust it more now that I know it can provide sources.” Respondent P9 noted that source provision during the task session was itself a positive and memorable experience, saying, “when it provides sources, I find that also helpful; the way it provided a peer-reviewed journal was nice.” Respondent P3 offered the most economical statement of the trust restoration mechanism: “because I know that I can find sources now, I would say my confidence goes up.”

In short, source transparency is not merely a feature preference. It determines whether students act on ChatGPT’s outputs at all. Without it, the full evaluative burden falls on the student with no means of acting on it. With it, students have a pathway to verify what they receive, making ChatGPT a more trustworthy and usable resource.

Trust Theme 8: Other Sources of Trust Failure. While source transparency is the primary mechanism through which trust may be built and restored, it does not fully account for why it fails in the first place. The mechanisms documented here are qualitatively different from a

sourcing deficit; they include fabricated information, limited evaluative capacity, and the weight students place on the consequences of acting on inaccurate outputs. Each operates independently of whether sources are provided, and each carries different implications for how trust might be addressed.

Fabricated and Hallucinated Information. The most consequential trust failure mechanism in the dataset was ChatGPT's documented tendency to generate fabricated information, particularly in academic contexts where source verification is both expected and necessary. What distinguishes this source of trust failure from general accuracy concerns is that participants were not expressing precautionary distrust; they were describing direct, witnessed encounters in which ChatGPT produced demonstrably false information. Respondent P4, who works as a research tutor, described the phenomenon saying:

In the academic field, [ChatGPT] makes up some sources, and you never know exactly 100% where they found that information. We call them ghost sources. AI would make up random sources. If you were to ask it to write a paper that has five academic sources about this topic, it would actually make up sources where the DOI didn't bring up anything in the system. It was just non-existent.

Respondent P8 described a parallel encounter in a professional library setting.

I had a lady come in, and she had a project due, and ChatGPT gave her all these articles, and she couldn't find them. I had to explain that it can fabricate—hallucinate. You need to go to Google after, and if you can't find it on Google or the library website, then it probably means ChatGPT made it up.

Where respondents P4 and P8 documented fabrication in professional contexts, Respondent P1 and Respondent P2 encountered it in the classroom. Respondent P1 described a

case in which ChatGPT's confident, well-formed response was simply wrong within a verifiable, closed-answer context: "I asked ChatGPT what it could be, and it gave me an answer, but the answer it gave me wasn't even on the list of possible choices. I had to go and find it in the study book instead." Respondent P2 described an experience that proved formative in shaping her distrust:

My teacher told us to try to get a generated citation for an essay. When she did it in front of the class, she tried to go back and find the sources, but they were made up. That experience definitely turned me around a little bit and made me feel like I can't trust it.

This account carries particular weight because the trust failure was not a private discovery but a public demonstration by an authority figure, one that scaled the damage across an entire cohort rather than a single user. Users expect the AI to be perfect, and when they experience a failure, this makes a deep impression, one that is hard to dispel.

Limited Familiarity as a Constraint on Evaluative Capacity. A second source of trust failure operated not at the level of the tools' outputs but at the level of the students' capacity to evaluate the factual accuracy of the results. Several participants described a form of trust failure rooted in their own limited familiarity with ChatGPT, not distrust of the tool per se, but an inability to judge whether trust was warranted. Respondent P2 located this explicitly in self-assessed competency, mentioning, "I'm not confident in my own abilities because I've rarely used it, so I know I wouldn't trust myself all too much." Respondent P8 described the same dynamic from a slightly different angle, noting that even after direct engagement with the tool, the pull of a more familiar and trusted alternative remained strong: "Inside I still felt like I needed to go back to Google because I trust Google more, so I'm not very confident yet." Respondent P9 pushed this further, identifying domain knowledge specifically as the variable

that determines whether evaluation is even possible, saying that, “If I can fine-tune it to things I’m somewhat familiar with, then it’s easier for me to determine how accurate it is. I would mainly use ChatGPT if I need a quick refresher on something I already know about.”

This formulation exposes a structural dependency. Without visible sources, accurate evaluation requires subject knowledge that the student may not yet have. This dependency compounds the fabrication mechanism documented above. Students with limited domain knowledge who are aware that AI may hallucinate are more suspicious of results, even when the output is accurate. For these students, evaluative capacity and source transparency are not separate concerns but converging ones.

Stakes-Calibrated Trust. A third source of trust failure emerged around the relationship between task stakes and trust investment. Participants described trust not as a fixed judgment but as something actively calibrated in response to the perceived consequences of acting on inaccurate information. Respondent P1 articulated this, saying, “If it were for a paper or assignment where I need credible sources, I’ll put more effort into it. If it’s for something not important, then I won’t put as much effort.” This account reveals that trust is not evaluated uniformly across all uses. Low-stakes queries tolerate lower levels of trust, while high-stakes academic tasks require a higher confidence threshold before a result is trusted. When that threshold is not met, the tool is either avoided altogether or restricted to low-stakes purposes.

Respondent P9 extended this stakes logic beyond academic contexts into real-world decision-making, offering the most expansive account of trust consequences in the dataset. On legal advice, respondent P9 noted his

mother is a lawyer concerned about how people will use AI for legal advice when AI is currently very unfit to provide that. Because we make these huge life decisions based on

the information that's provided for us, it's really important that it's well-sourced and that we can look into it.

The same concern applied to medical contexts:

If somebody is experiencing symptoms like sadness, they may look that up, and AI may tell them they have depression. If the person were well-equipped, they would be able to evaluate the information. But if they aren't, they may jump to conclusions when they should be talking to a doctor.

Taken together, these accounts position trust evaluation as a fundamentally consequentialist process. The greater the stakes attached to acting on information, the higher the trust required before that information is accepted. For ChatGPT, where trust is already hedged across the sample, high-stakes contexts represent an acute point of failure.

These three sources of trust failure illustrate why the trustworthiness item produced the most one-directional result in the dataset. Students are not simply less familiar with ChatGPT than with Google; they have specific, documented reasons for mistrust, often have limited evaluative capacity to verify results, and, when queries have high-stakes consequences, are less likely to trust the results. These mechanisms do not operate independently; they compound. A student with limited domain knowledge who encounters a fabricated source in a high-stakes academic context faces a convergence of trust failure conditions that source transparency alone cannot resolve. The data also reveal that when trust fails, students do not simply stop using ChatGPT; they develop a behavioral workaround.

Information Retrieval Behavioral Comparison

The findings and analysis from the perception section established that students experience trust concerns with both Google and ChatGPT as information retrieval tools, though

the sources of those concerns differ. When using Google, students described SEO-driven obstacles to trust, including prioritizing sponsored content and issues with the relevance and credibility of results. When using ChatGPT, students reported a lack of source transparency, the risk of fabricated or hallucinated information, difficulty assessing response correctness due to limited familiarity with the content, and the verification burden that low trust imposes. This section examines how students navigate those concerns behaviorally during actual information retrieval and what those behaviors reveal about how each tool structures the work of establishing trust.

In order to answer Research Question 2, participants completed five information retrieval tasks using Google Search and ChatGPT in separate sessions. Table 4 presents task completion times and query attempt counts across the five tasks. ChatGPT completed tasks faster than Google on every task, with the largest gaps on Task 4 (resources) and Task 5 (fact-checking). Query counts between the two tools were similar: overall users averaged 2.3 Google queries to complete the five tasks and 1.9 ChatGPT prompts. However, users required twice as many queries to complete Task 5 on Google as on ChatGPT.

Table 4

Task Time & Query Paired Comparisons

Task	Google Time	ChatGPT Time	Google Queries	ChatGPT Prompts
T1 Basic information	2:57	2:10	2	1.7
T2 Complex information	3:26	2:02	1.5	1.3
T3 Instructions	2:58	2:10	1.5	1.6
T4 Resources	5:34	4:28	2.2	2.4
T5 Fact checking	6:28	4:00	4.4	2.8

Note. Time in minutes and seconds.

Time and query data confirm that students moved through ChatGPT tasks more quickly, but they do not explain what drove that difference. Tasks requiring a direct answer (T1–T3) clustered around 3 minutes on Google; tasks requiring source evaluation or verification (T4–T5) took roughly twice as long. Since query counts were comparable across tools while completion times were not, the gap reflects where effort was concentrated. On Google, effort was invested in vetting results (i.e., scanning, comparing sources, and assessing credibility). On ChatGPT, effort was spent constructing and refining prompts, since prompting is the only form of interaction the tool affords. In both cases, the single largest driver of behavioral variation and completion time was how students responded when a tool failed to earn their trust.

Observations of the search activity revealed specific behaviors students took to mediate trust issues. It should be noted that when conducting an information retrieval task with Google, the tool often, but not always, provides an AI-generated response similar to ChatGPT. This was often accompanied by a list of potential sources that could be used to answer the query. Sponsored sources were listed before organic sources based on the tools' SEO algorithm. At the time of this study, Google AI summaries did not automatically provide links to the sources used to create the response. ChatGPT, on the other hand, provides the most probable answer to the prompt based on patterns it had encountered during training. It may also search the web to find up-to-date information, which it summarizes and includes in a response. At the time of this study, ChatGPT was using OpenAI's 4.0 mini model.

A central finding from observations was that, for both tools, users ended their searches when they trusted the answers they received. Figure 1 shows the hierarchical effort users put into completing a task using Google. Each level describes a point at which the user found a trusted result and ended the search. Each level requires more effort in verifying trust than the last. Figure

2 shows the cyclical effort users put into completing a task using ChatGPT. Unlike Google, ChatGPT offers no hierarchy of levels because effort is concentrated entirely in prompting. Each trust or comprehension failure returns the user to another prompt, and this loop continues until the user trusts the output or exits the tool to verify on Google.

Figure 1

Google's Effort and Trust Hierarchy

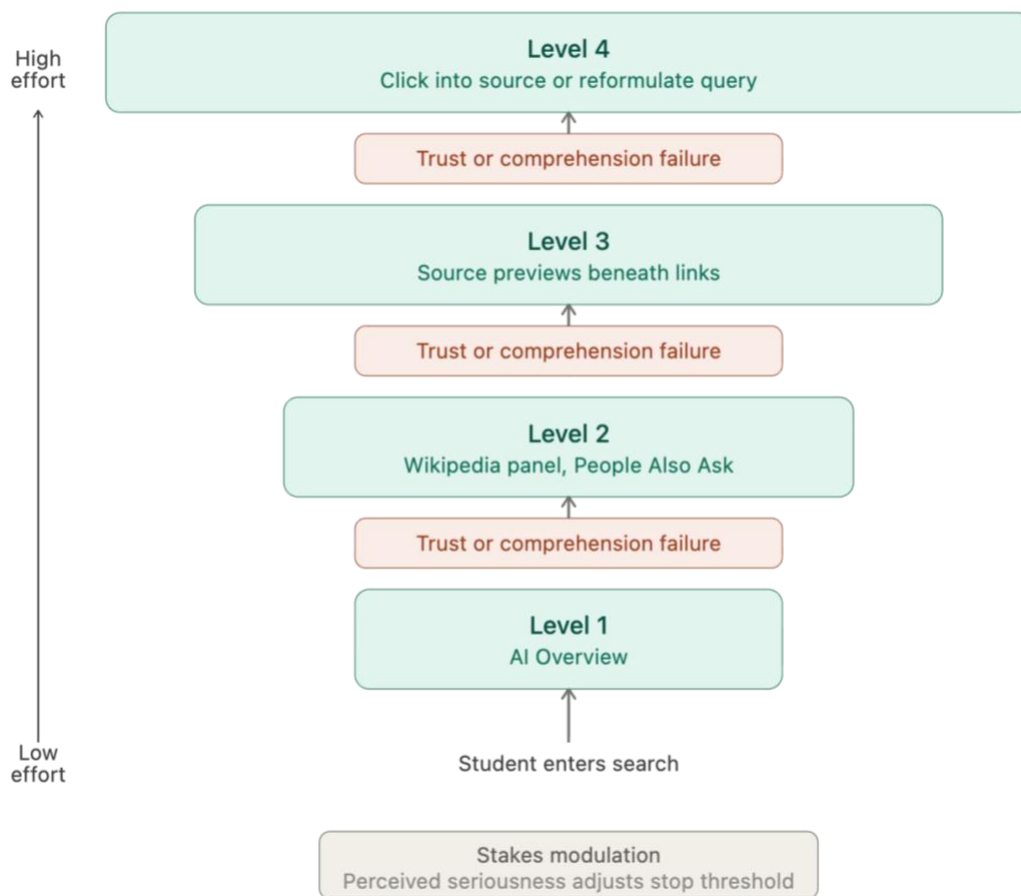
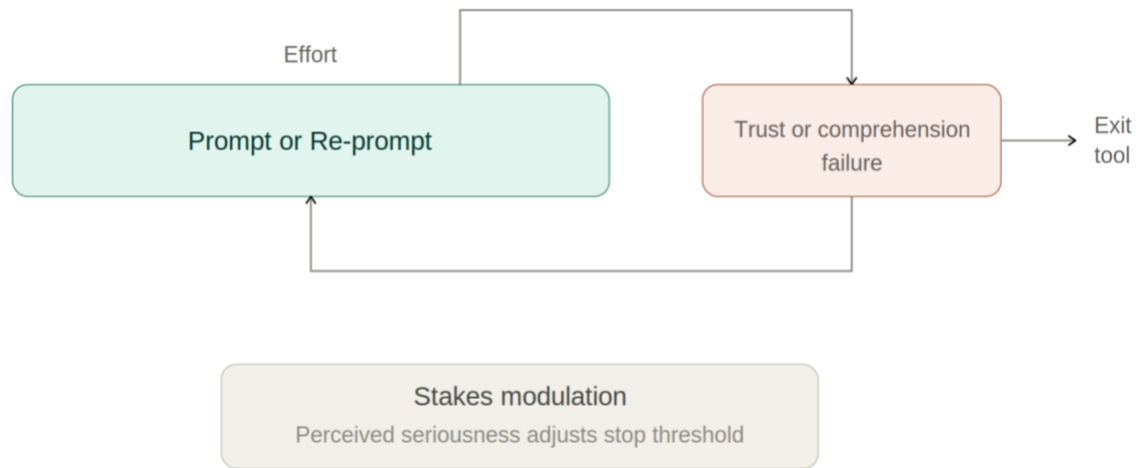


Figure 2*ChatGPT's Effort and Trust Loop****Task Importance and Trust Mediation***

It was noted that students calibrate their trust thresholds to the consequences of acting on inaccurate information. This calibration operated behaviorally across both tools and across all the behaviors described below. On Google, stakes influenced how much effort they put into completing an information retrieval task. Low-stakes queries often resolve at the AI Overview, while high-stakes tasks push students into source previews and full source reviews. On ChatGPT, stakes influence whether students accept the initial response, re-prompt for verification, or exit the tool entirely to verify on Google. Stakes do not change which trust behaviors are available in either environment; they change how thoroughly students deploy them.

Google's Trust Resolution

Four trust-resolution behaviors emerged across observations, each representing a different strategy for converting uncertainty into a workable level of trust.

Cross-Source Triangulation. The most widespread trust-resolution behavior when using Google was cross-source corroboration. When students encountered the same information across multiple independent sources, they described increased confidence and stopped searching. For example, respondent P8 on the fact-checking task noted that consistency across the People Also Ask, Wikipedia, and Quora sources gave her the confidence to end her search. Respondent P2 on the basic information task described the underlying logic directly stating, “If all of the sources say this is his first novel, it makes me trust that it’s his first novel.” P7 on the same task verified a single fact across three sources (i.e, the AI Overview, Wikipedia, and Penguin Random House) before accepting it. Triangulation operated at every level of the hierarchy; students could compare the AI Overview against Wikipedia or People Also Ask at Levels 1 and 2, scan source previews at Level 3 for repeated claims, or cross-reference full sources at Level 4. This flexibility made triangulation the resolution behavior most compatible with low-effort navigation, and it appeared across all five tasks.

Domain Evaluation. Students used the URL domains as a low-effort trust shortcut. Educational, government, and organizational domains built confidence and allowed students to stop at a source without further investigation. For example, respondent P2 on the fact-checking task stated simply, “This source is from Yale, so I trust it.” Respondent P10 on the instructions task explained her selection of sources, stating, “I’m going to look for a source that is from a college because colleges do a lot of research and have been checked.” Respondent P4 extended this trust logic to government sources, noting that she trusted .gov sources for basic informational content even while acknowledging that government sources are not uniformly reliable across all topics. In this sense, a domain evaluation functioned as a heuristic that let students bypass the more effortful resolution behaviors, as they didn’t need to investigate source

authority if the domain itself already signaled credibility.

Source Familiarity. A third trust resolution behavior relied on prior experience with a source. Students selected and trusted sources they had personally used before, encountered in a previous class, or recognized as a familiar name. For example, respondent P8 on the instructions task trusted the Cleveland Clinic because “I’ve used Cleveland Clinic before.” Respondent P2 trusted Litcharts from her Shakespeare class and Britannica because her high school had “relied on that source a lot and told us it was a very trustworthy source.” Respondent P9 selected Goodreads on the basic information task because “I’ve used that in the past” and knew exactly where to navigate on the site. Familiarity served as another low-effort shortcut, strong enough to override domain-based hesitation.

Source Authority Checking. When using Google, the most effortful resolution behavior involved direct investigation of a source’s authority, including evaluating its institutional affiliation, author credentials, or publication history. Respondent P4 was the most systematic source verifier when using Google for Task 5; she indicated that, before cross-referencing sites to confirm their credibility, “I would look up these websites to check their source authority.” Respondent P10 demonstrated a multi-step verification process on the same task, moving from content evaluation to publisher verification to author credibility. Respondent P2 on the instructions task applied a simpler version of the same logic, trusting a medical source because “a doctor knows what they are talking about.” Authority checking often required students to leave the results page, conduct additional searches, and evaluate what they found.

ChatGPT Trust Resolution

The behavioral structure that emerged from the ChatGPT sessions is qualitatively different from what emerged on Google. Google offered students a hierarchy of information

sources within a single results page, with each level requiring more effort than the last and each supporting different trust-resolution behaviors. ChatGPT offers no such hierarchy. A single prompt produces a singly synthesized response, and there are no intermediate levels of engagement between reading that response and doing something about it. The only form of effort available to the students inside the tool is prompting.

This structural difference concentrates student effort at two moments in the interaction. First, at the initial prompt, where students invest in shaping the response before it arrives. Second, at subsequent prompts, which students submit when the initial response triggers a trust failure. Together, these two moments account for most of the behavioral variation observed in the ChatGPT sessions.

Initial Prompt Effort. Students who invested effort in their initial prompt did so to front-load trust, shaping the output before it arrived rather than correcting it afterward. Two techniques emerged across the sample as trust-preventive strategies.

Anti-hallucination prompts explicitly instructed ChatGPT not to fabricate information (e.g., “do not make up sources,” “only use real citations”). Students who used this approach described doing so because they had encountered fabricated output in the past or had been warned by instructors, classmates, or coworkers. It is the most direct form of front-end trust work; the student anticipated potential trust failure and attempted to engineer around it before the response was generated.

Role assignment instructed ChatGPT to adopt a specific persona or area of expertise (e.g., “act as a research librarian,” “respond as if you are a historian”), orienting the response toward a recognized type of authority. Respondent P8 described this as a learned strategy, explaining that she had learned to “give it a role and a target ... something it needs to accomplish and the

specific actions it needs to do.” Unlike anti-hallucination prompts, role assignment does not directly address fabrication. Instead, it shapes the response’s credibility register by cueing the tool toward an authoritative voice.

Re-Prompting Effort. When the initial ChatGPT response failed to meet a student’s trust threshold, the only recourse available inside ChatGPT was to issue another prompt. Where initial prompt effort is preventive, re-prompting is reactive, triggered by something in the response that the student could not accept without resolution. Two re-prompting behaviors emerged across the sample as trust-resolution moves.

Internal repetition was the more frequent of the two. Students asked ChatGPT the same question in a different form, or asked the tool to verify its own previous answer, treating consistency across prompts as a proxy for accuracy. Respondent P9 described the behavior on Task 1, saying, “just to make sure that these parts right here aren’t just kind of supplementary, I’m going to clarify and prompt again.” Respondent P2 described the same strategy explicitly as a verification move, saying, “I would immediately want to ask another question that’s the same thing reworded so that I can just double check that it spits out the same information ... it’s just easier to flip it around and double-check.” Respondent P8 used a related approach, indicating, “I know things can be wrong, so I sometimes will ask it to check itself.” Internal repetition illustrates the constraint of ChatGPT’s single-lever structure: when accuracy is in doubt, prompting again is the only path available.

Source requesting was the most direct trust-resolution move available inside the tool. When the initial response did not include citations, students re-prompted to ask for them. Respondent P5 described this as a trust move, saying, “I really just don’t know how to believe the information I get on here, so I will try to find out where it comes from.” The presence of

sources generally increased trust, as students took their inclusion as a signal that the response was grounded in real information rather than generated without reference. Source requests, however, frequently marked the boundary of ChatGPT's usefulness. Students who received citations from ChatGPT often left the tool to verify that those sources actually existed using Google.

Taken together, the prompting behaviors described above appeared less consistently across the sample. Where Google's resolution behaviors were deployed routinely by most participants across most tasks, ChatGPT's trust-preventive and trust-resolution prompting behaviors appeared in smaller subsets of the sample and were deployed more variably from task to task. This is reasonably explained by the fact that participants report being less familiar with using ChatGPT than Google.

When Trust Cannot Be Resolved with ChatGPT. For many participants in this study, the trust-resolution behaviors available in ChatGPT had a ceiling effect. Asking ChatGPT to check the response or provide sources did not always overcome respondents' skepticism about the accuracy of AI in general. When re-prompting could not resolve a trust concern, students often chose to exit ChatGPT and verified on Google.

Several participants described this strategy. For example, respondent P4 captured its most extreme reaction, abandonment: "I'm not 100% confident in its information, which means I just have to go to Google to check anyway. So I would just skip ChatGPT altogether, unless I'm brainstorming." Respondent P5 described how the verification step erased ChatGPT's speed advantage, saying, "if I'm trying to find valid sources for school, then it takes longer than Google. It's adding an extra step because I end up on Google anyway." Respondent P1 normalized the exit as routine rather than exceptional, saying, "you can go and study to see if

ChatGPT's answer is correct, same thing you have to do on Google.” Together, these accounts point to a practical ceiling on ChatGPT's utility for academic tasks. Until users gain trust in AI as a reliable, accurate source, some will choose not to use it.

Limitations

This study has four limitations that should be considered when interpreting its findings. First, data were collected between late 2024 and early 2025, a period in which generative AI tools were evolving rapidly. Participants' perceptions and behaviors may reflect features, affordances, and constraints specific to that moment, and some findings may not hold as these tools continue to develop.

Second, the study involved a small, purposefully selected sample of 10 undergraduate students from a single university. The sample was appropriate for the depth of qualitative analysis required by this study, but the findings should not be generalized to undergraduate populations without further investigation.

Third, many participants reported limited prior experience with ChatGPT, which means the perceptions and behaviors documented here may reflect early-stage familiarity rather than more stable patterns that develop with sustained use or greater prompt literacy. How students engage with generative AI tools is likely to shift as proficiency increases.

Fourth, thematic analysis requires interpretive judgment, and the researcher's perspective inevitably shapes how data are coded and findings are constructed. Reflexive awareness was maintained throughout the analytical process, but the possibility of interpretive bias cannot be eliminated.

Conclusion

This study examined undergraduate students' information-retrieval behaviors and

perceptions when using a generative AI tool (i.e., ChatGPT) compared to a traditional keyword-based search engine (i.e., Google). Across both the perception and behavioral data, six findings emerged as central.

First, trust is the gating variable for how students engage with generative AI in academic information retrieval. The technology acceptance model, as formulated by Davis (1989) and extended by Gefen et al. (2003), positions perceived ease of use, perceived usefulness, and trust as three coequal predictors of technology adoption. The present data suggest a different relationship. Trust does not operate alongside the other two constructs; it operates as a foundational element that determines perceived ease of use and usefulness. When students trusted ChatGPT's AI-generated responses, they realized the tool's advantages, including synthesized responses and the ability to engage conversationally with the AI. When trust was insufficient, those advantages collapsed with the cost of verification offsetting convenience and speed. Across TAM construct analyses, trust determined whether ease of use and usefulness carried practical weight, a pattern that suggests a better model of trust as a moderator of those constructs than as a parallel variable.

Second, source transparency was the primary mechanism through which trust was regulated. It was the most consistently reported theme in the dataset, described in some form by all 10 participants. Its absence left students with no pathway for independent verification, eroding confidence in ChatGPT's outputs. Its presence restored trust, with several students reporting a meaningful shift in their perception of the tool after discovering it could produce sources on request. Students' confidence depended less on the actual accuracy of an output than on whether a verification pathway was available.

Third, a stakes evaluation calibrated how much trust work students required. The

perception data showed that students adjust their trust thresholds based on the consequences of acting on inaccurate information, and this calibration carried over into their behavior across both tools. Low-stakes tasks tolerated lower levels of verification and acceptance. High-stakes tasks, particularly graded assignments, raised the trust threshold and pushed students to invest additional effort before accepting what they found.

Fourth, the two tools constitute structurally different trust environments, and that difference is a property of the tools rather than of the students using them. Google offers a hierarchy of trust-resolution behaviors (e.g., triangulation, domain evaluation, source familiarity, and authority checking) that can be deployed at varying levels of effort. ChatGPT offers no such hierarchy. A single prompt produces a single synthesized response, and the only form of effort available inside the tool is re-prompting. When prompting cannot resolve a trust concern, students often resort to Google to verify results—a tool they are familiar with. This structural asymmetry shapes how students manage trust in each environment and accounts for much of the behavioral variation documented in the Results.

Fifth, students' Google trust behaviors are settled and routine, while their ChatGPT trust behaviors are emergent and experimental. Most participants consistently deployed trust verification mechanisms when using Google. This was likely the result of years of schooling and daily use. By contrast, trust-prevention and trust-resolution behaviors used with ChatGPT were used by fewer students in this sample and were deployed more variably across tasks. Only a few participants named what they were doing as prompt engineering, and most described arriving at these techniques through trial and error rather than formal instruction, suggesting that students are only beginning to develop AI trust and prompt literacy.

Lastly, adoption of ChatGPT continues to accelerate despite the trust deficit documented

in this sample, a pattern consistent with broader data on rapid AI adoption among students (Digital Education Council, 2024; Sidoti et al., 2025). Two factors appear to sustain this trajectory. The forms of usefulness ChatGPT offers (i.e., speed, synthesis, and conversational depth) are accessible early in the adoption curve and were recognized even by participants who self-identified as beginners. Additionally, the perceived costs of non-adoption are real. If ChatGPT makes academic work faster and more efficient, students who do not engage with it risk falling behind peers who do, particularly as AI competence becomes an increasingly explicit professional expectation. This combination, early accessible advantage and competitive pressure, sustained in the presence of incomplete trust, bears structural resemblance to what Sunstein describes as a product trap (i.e., situations in which continued use is driven by external dependencies rather than individual satisfaction). The present study does not claim that ChatGPT use constitutes a product trap in the formal sense defined by Bursztyn et al. (2025), but the framework offers a productive lens for understanding why adoption continues to grow even as trust remains underdeveloped.

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APPENDIX A

Annotated Bibliography

Introduction

This annotated bibliography supports the thesis, *Exploring Generative Information Retrieval Behaviors Among Students: An Investigation of Google Search and ChatGPT Queries*, which examines how undergraduate students perceive and behaviorally engage with ChatGPT and Google Search as academic information retrieval tools. The thesis is grounded in the Technology Acceptance Model (TAM; Davis, 1989), extended to include trust (Gefen et al., 2003), and employs an embedded mixed-methods design combining Likert-scale perception ratings with thematic analysis of semi-structured interviews, structured information-retrieval tasks, and concurrent think-aloud protocols.

The annotated bibliography spans four interconnected areas of scholarships: (a) foundation theories of information seeking and retrieval, which establish the behavioral and cognitive frameworks against which students' AI-mediated practices are interpreted; (b) research on search engines and generative AI, which documents the emerging behavioral patterns associated with AI-mediated information retrieval; (c) technology acceptance and adoption frameworks, which provide the theoretical architecture for understanding why students use (or resist) new information tools; and (d) trust and perception in AI, which examines the dynamic relationship between user confidence, source transparency, and continued tool use. Forty-four sources are included, and the scope of the bibliography is intentionally broad enough to establish the context for the research questions while remaining focused on sources that bear directly on the study's framework, methods, findings, or interpretive conclusions.

Literature Search Methodology

Databases and Publication Outlets

Sources were identified through searches of the following databases and publication outlets: EBSCO Academic Search Complete, JSTOR, Web of Science, Google Scholar, ProQuest Dissertations and Theses, ACM Digital Library, and the ERIC (Education Resources Information Center) database. Targeted searches were also conducted in high-impact journals relevant to the study's disciplinary domains, including *MIS Quarterly*, *Journal of Documentation*, *British Journal of Educational Technology*, *Computers and Education*, *Journal of the American Society for Information Science*, *International Journal of Information Management*, and *Qualitative Research in Psychology*. For practitioner and technology-oriented sources, searches included Nielsen Norman Group website, the Pew Research Center, the Digital Education Council, the World Happiness Report, and Technology from Sage. The arXiv preprint server was used to access recent empirical work on AI-mediated information behavior not yet available through peer-reviewed channels.

Search Strategies and Time Period

Searches were conducted iteratively across the Fall of 2024 through Spring 2026 data collection period, with foundational sources identified first and more recent empirical work added as it became available. Two categories of search strategy were employed. For the foundational literature on information seeking, technology acceptance, and qualitative research methods, searches use canonical author names and model titles as primary terms (e.g., “Technology Acceptance Model Davis,” “Kuhlthau information search process,” “Braun Clarke thematic analysis,” “Dervin sense-making”). These searches were not time-bounded, as the scholarly significance of these sources does not diminish with age.

For the empirical literature on generative AI in education and student information-retrieval behavior, searches used keyword combinations drawn from the thesis's research questions and theoretical constructs. Primary keyword clusters included the following: (a) generative AI + students + information retrieval; (b) ChatGPT + undergraduate + academic; (c) trust + artificial intelligence + user behavior; (d) technology acceptance + AI + higher education; (e) prompt engineering + search literacy; (f) AI adoption + education; and (g) information-seeking behavior + search engine. The searches that produced publications from 2020 onward were considered, with sources published between 2022 and 2026 being prioritized to capture the most current empirical findings in a quickly evolving field. Earlier sources were only included if they provided foundational context or comparative data.

Inclusion Criteria

1. Relevance to framework or methods. Sources were included if they directly informed the study's theoretical framework (TAM, trust extension, information-seeking models), research methodology (embedded mixed-methods design, thematic analysis, think-aloud protocols, purposeful sampling, scale design), or interpretive conclusions (product trap framework, AI hallucination, prompt engineering).
2. Empirical or scholarly grounding. Peer-reviewed journal articles and conference proceedings were prioritized. Books, book chapters, and methodological handbooks were included when they served as primary methodological authorities. Practitioner reports, industry white papers, and government or institutional surveys were included when they provided empirical data not available through peer-reviewed channels (e.g., adoption rate statistics or global survey data).
3. Disciplinary scope. Sources were drawn from information science, educational

technology, human-computer interaction, qualitative research methodology, and organizational behavior. Interdisciplinary or practitioner-facing sources were included when they addressed empirical questions directly relevant to the thesis.

4. Active citation in the thesis. All sources included in this annotated bibliography are cited in the thesis itself. Sources consulted during the literature review but not directly cited in the final text were excluded.

Category 1: Foundational Theories of Information Seeking and Retrieval

The foundational models collected here (Wilson [1981, 1999], Dervin [1983, 1992], Ellis [1989], Ellis et al. [1993], Kuhlthau [1991], and Fadaie [2012]) collectively established that information seeking is an iterative, cognitive, and affective process shaped by individual context and available tools, not a simple transactional exchange with a retrieval system. Empirical work by Dinet et al. (2012) and Civilcharran et al. (2015) confirmed that effective searching is a developable, context-sensitive skill and that students gravitate toward familiar, low-effort tools. Sundin et al. (2017), Coetzee and Stevens (2025), and Technology from Sage (2023) document the long-term cultural normalization of Google and the broader shift from library-based to internet-based research, contextualizing AI-mediated retrieval as the next stage in that transformation.

Civilcharran, S., Hughes, M., & Maharaj, M. S. (2015). Uncovering web search tactics in South African higher education. *South African Journal of Information Management*, 17(1), 1–8.

This qualitative investigation examined the web search strategies employed by undergraduate students in a South African university, finding that students overwhelmingly favored familiar, accessible tools and relied on heuristic rather than systematic source evaluation. Students tended to prioritize speed and perceived ease of

use over scholarly rigor. For the thesis, this source supports the claim that student information-retrieval behavior is shaped by tool familiarity and low-effort heuristics, providing cross-national evidence for a pattern also documented in the thesis sample and reinforcing the TAM constructs of PEOU and trust as key adoption drivers.

Coetzee, N. J., & Stevens, K. (2025). When the library closes: Assessing the academic impact on engagement, citation practices, and performance in first-year students. *Library & Information Science Research*, 47(3), Article 101399.

<https://doi.org/10.1016/j.lisr.2025.101399>

This study examined the impact of reduced library access on first-year university students' academic engagement and citation practices, finding that students increasingly defaulted to internet-based tools (particularly search engines) when physical library resources were unavailable. Citation quality and source diversity declined in this context. For the thesis, this source substantiates the introductory claim that digital information retrieval has displaced traditional library-based research and that search engines have become the dominant gateway to academic information, positioning the shift to AI-mediated retrieval as the next stage in this longer-term trend.

Dervin, B. (1983, May). *An overview of sense-making research: Concepts, methods, and results to date* [Paper presentation]. International Communication Association Annual Meeting, Dallas, TX, United States.

In this early theoretical paper, Dervin introduced the sense-making methodology as a framework for understanding how individuals actively construct meaning when navigating information gaps. The framework positions information retrieval as a cognitive and situational process shaped by perceived gaps between what is known and

what is needed, rather than as a simple transactional exchange with a retrieval system.

For the thesis, Dervin's model is cited as one of several foundational frameworks in the information-retrieval literature that collectively establish the conceptual heritage against which AI-mediated retrieval behaviors must be interpreted.

Dervin, B. (1992). From the mind's eye of the user: The sense-making qualitative-quantitative methodology. In J. D. Glazier & R. R. Powell (Eds.), *Qualitative research in information management* (pp. 61–84). Libraries Unlimited.

This chapter elaborates on the sense-making methodology and its application in information management research, emphasizing the need to study information-retrieval from the user's subjective perspective rather than from the standpoint of system design or content retrieval. Dervin argues for mixed-method approaches that integrate quantitative and qualitative data to capture the full complexity of how users construct and navigate knowledge gaps. This source informs the thesis's methodological orientation and its attention to participants' subjective accounts of tool perception, reinforcing the decision to foreground qualitative interview data over quantitative Likert ratings.

Dinet, J., Chevalier, A., & Tricot, A. (2012). Information search activity: An overview. *European Review of Applied Psychology*, 62(2), 49–62.

This comprehensive review surveys decades of empirical research on information search behavior, synthesizing findings on how users formulate queries, evaluate relevance, and navigate digital information environments. The authors document that effective search is a developable skill and that students who lack strategic search proficiency tend to retrieve lower-quality information. For the thesis, this source grounds the argument that internet search literacy is a foundational academic competency, providing the disciplinary context

for the thesis's later claim that prompt engineering may be emerging as an analogous but distinct competency for AI-mediated retrieval.

Ellis, D. (1989). A behavioural approach to information retrieval system design. *Journal of Documentation*, 45(3), 171–212. <https://doi.org/10.1108/eb026843>

This study developed one of the earliest empirically grounded behavioral models of information seeking, identifying a set of characteristic activities (including starting, chaining, browsing, differentiating, monitoring, and extracting) that researchers employ across information-retrieval tasks. Ellis's model shifted attention from system architecture to user behavior, establishing a behavioral tradition in information retrieval research. For the thesis, this source is cited as part of the foundational literature on information-seeking models, establishing the disciplinary context in which the present study's behavioral findings (trust-resolution hierarchies on Google, prompting loops on ChatGPT) can be situated.

Ellis, D., Cox, D., & Hall, K. (1993). A comparison of the information-seeking patterns of researchers in the physical and social sciences. *Journal of Documentation*, 49(4), 356–369. <https://doi.org/10.1108/eb026919>

This comparative study extended Ellis's (1989) behavioral model across two academic disciplines, examining whether information-seeking patterns differed between physical and social science researchers. The authors found broadly consistent behavioral patterns but documented meaningful disciplinary variation in how researchers chained references and differentiated sources. For the thesis, this source contributes to the established behavioral tradition in information retrieval research and supports the argument that information-seeking patterns are shaped not only by individual disposition but by the

tools, contexts, and norms available to users.

Fadaie, G. (2012). Information seeking, information retrieval: Philosophical points. In *Proceedings of InSITE 2012: Informing Science + IT Education Conference* (pp. 121–128). Informing Science Institute. <https://doi.org/10.28945/1643>

This theoretical paper proposes a typology of information seekers based on the nature of their information needs and retrieval strategies, distinguishing among exploratory, confirmatory, and serendipitous seeking styles. Fadaie situates information seeking as a broader, more cognitively complex process than the mechanical retrieval that characterizes algorithmic search. For the thesis, this source contributes to the distinction between information retrieval (IR) and information seeking (IS) introduced in the literature review, supporting the argument that AI-mediated tools must be evaluated not only for retrieval efficiency but for how well they support the more exploratory cognitive processes students bring to academic research.

Kuhlthau, C. C. (1991). Inside the search process: Information seeking from the user's perspective. *Journal of the American Society for Information Science*, 42(5), 361–371.

Kuhlthau's Information Search Process (ISP) model describes information seeking as an affective and cognitive journey across six stages (initiation, selection, exploration, formulation, collection, and presentation) with associated feelings of uncertainty, optimism, confusion, and confidence. The model situates information seeking as a process of constructing understanding rather than simply locating facts. For the thesis, the ISP model is cited as a foundational framework in the information-seeking literature and provides context for interpreting students' affective responses (uncertainty, frustration, increased confidence) during ChatGPT and Google tasks.

Sundin, O., Haider, J., Andersson, C., Carlsson, H., & Kjellberg, S. (2017). The search-ification of everyday life and the mundane-ification of search. *Journal of Documentation*, 73(2), 224–243.

This theoretical paper examines the cultural and institutional normalization of search engine use, arguing that Google has become so embedded in everyday life and educational practice that searching has become "mundane"—a taken-for-granted cognitive habit rather than a deliberate literacy skill. The authors trace how educational institutions incorporated search literacy in direct response to Google's ubiquity. For the thesis, this source supports the argument that students' high PEOU ratings for Google reflect decades of integration into academic and social technology environments, and provides historical and sociological context for why Google is perceived as familiar and effortless while ChatGPT is perceived as requiring effortful skill development.

Technology from Sage. (2023). *The knowledge gap between librarians and students:*

Contrasting librarian and student perspectives on the undergraduate workflow

(Librarian Futures, Part 2). https://www.technologyfromsage.com/wp-content/uploads/2024/09/LibrarianFutures_Part2_FINAL.pdf

This industry report from a scholarly communications technology company surveys both academic librarians and undergraduate students about research workflows and information-seeking behaviors. The report documents a well-established shift away from physical library resources toward internet-based tools, with students predominantly relying on search engines and increasingly on AI tools for academic research. For the thesis, this source provides the introductory evidence that digital search has displaced traditional library-based research, contextualizing the thesis's study of AI-mediated

information retrieval as the continuation of a broader and well-documented transformation in how students access academic knowledge.

Wilson, T. D. (1981). On user studies and information needs. *Journal of Documentation*, 37(1), 3–15.

In this foundational article, Wilson argued that information retrieval research must be grounded in an understanding of users' information needs and the social, environmental, and personal contexts that shape those needs. He proposed a broad conceptual framework for understanding the relationship between information needs, information-seeking behavior, and information use, insisting that the user (not the system) must be the unit of analysis. For the thesis, Wilson's model is cited as one of the foundational frameworks in the information-seeking tradition, establishing the user-centered orientation that informs the thesis's methodological approach and its emphasis on students' subjective accounts of their tool use.

Wilson, T. D. (1999). Models in information behaviour research. *Journal of Documentation*, 55(3), 249–270.

In this retrospective review, Wilson surveyed and compared the major theoretical models in information behavior research, including his own 1981 framework and those proposed by Dervin, Ellis, and Kuhlthau. He critically assessed the empirical and conceptual foundations of each model and identified recurring themes, including the iterative and context-dependent nature of information seeking and the central role of cognitive and emotional states in shaping retrieval behavior. For the thesis, this source is cited alongside the other foundational information-behavior models to establish that information seeking has long been recognized as a complex, user-centered process, one

that AI-mediated tools now disrupt in ways that require systematic empirical investigation.

Category 2: Search Engines and Generative AI in Academic Contexts

Sources in this category documented how generative AI differs from traditional keyword-based search and how those differences show up in user behavior. Lund and Wang (2023) established that generative AI collapses the search-and-review distinction by synthesizing information into conversational responses rather than returning ranked links. Schuetzler et al. (2024) and Gibbons et al. (2023) each documented distinct AI interaction patterns (“prompt cycling” and efficiency-driven summary retrieval) that reflect a prioritization of cognitive ease distinct from traditional search behaviors. Mayerhofer et al. (2025) documented tool-switching and trust-conditioned verification practices that parallel the ChaGPT trust loop observed in the thesis. Together, these sources establish that AI-mediated retrieval requires a qualitatively different skill set, centered on prompt design rather than keyword construction.

Gibbons, S., Mugunthan, V., & Nielsen, J. (2023). *Accordion editing and apple picking: Early generative-AI user behaviors*. Nielsen Norman Group.

<https://www.nngroup.com/articles/accordion-editing-apple-picking/>

This UX research report from the Nielsen Norman Group documents early behavioral patterns in users’ interactions with generative AI tools, identifying two distinctive tendencies: accordion editing (iteratively expanding and collapsing AI outputs to refine content) and apple picking (selectively harvesting usable pieces of AI-generated text).

These patterns reflect a prioritization of cognitive ease and efficiency that is consistent with but behaviorally distinct from traditional search behaviors. For the thesis, this source contributes to the literature review’s characterization of AI-specific retrieval behaviors

and supports the argument that students using ChatGPT engage in qualitatively different interaction patterns than those documented in keyword-based search research.

Lund, B. D., & Wang, T. (2023). Chatting about ChatGPT: How may AI and GPT impact academia and libraries? *Library Hi Tech News*, 40(3), 26–29.

<https://doi.org/10.1108/LHTN-01-2023-0009>

This brief but frequently cited commentary piece examines the implications of ChatGPT for academic libraries and scholarly information practices, noting that generative AI collapses the traditional distinction between searching and reviewing by generating synthesized responses rather than returning ranked links. The authors flag implications for information literacy, source evaluation, and the role of librarians as mediators of research quality. For the thesis, Lund and Wang are cited at the transition from traditional search to AI-mediated retrieval in the introduction, providing disciplinary grounding for the claim that prompt generation and information synthesis are replacing keyword search literacy as core information retrieval skills.

Mayerhofer, K., Capra, R., & Elweiler, D. (2025). *Blending queries and conversations:*

Understanding tactics, trust, verification, and system choice in web search and chat interactions. arXiv. <https://arxiv.org/abs/2504.05156>

This empirical study compared user behaviors during web search and chat-based AI interactions, finding that users often accepted AI-generated responses without verification when outputs appeared well-formatted and plausible, even when they acknowledged uncertainty about accuracy. The study also documented patterns of tool-switching, with users returning to traditional search to verify AI outputs. For the thesis, this source provides direct empirical support for two central findings: that source transparency is the

primary mechanism through which trust is regulated, and that ChatGPT's fluent synthesis can paradoxically undermine informed trust by presenting outputs with no visible verification pathway.

Schuetzler, R. M., Giboney, J. S., Wells, T., Richardson, B., Meservy, T. O., Sutton, C., Posey, C., Steffen, J., & Hughes, A. (2024). Student interaction with generative AI: An exploration of an emergent information-search process. *Proceedings of the 57th Hawaii International Conference on System Sciences*. <https://doi.org/10.24251/HICSS.2024.901>

This conference study investigated how students interact with generative AI during information-seeking tasks, documenting a pattern the authors term "prompt cycling" (the iterative refinement of queries through conversational exchange) as a distinctive and fundamentally different process from discrete keyword-based search. The study provides an early empirical account of the behavioral mechanics of AI-mediated retrieval. For the thesis, Schuetzler et al. are cited in the literature review on search engines and generative AI to establish that the behavioral differences between traditional search and AI tools are already observable and documented, and to position the thesis's investigation within an emerging body of empirical work.

Category 3: Technology Acceptance and Adoption

Davis (1989) and Gefen et al. (2003) form the two theoretical pillars of this category. Davis introduced TAM's core constructs of perceived usefulness and perceived ease of use, while Gefen et al. extended TAM to include trust for contexts where users rely on system-generated outputs they cannot independently verify. Granić and Marangunić (2019) validate TAM's explanatory power across education technology contexts and confirm that trust is its most common extension. The remaining sources document the student AI adoption landscape.

Digital Education Council (2024), Sidoti et al. (2025), and Gillespie et al. (2025) establish the global scale and pace of adoption, while Bushweller (2024), Lambert and Stevens (2023), Tindle et al. (2023), Dwivedi et al. (2023), Yusuf et al. (2024), and Baek et al. (2024) collectively document how institutional responses have focused on restriction over empirical understanding. Bursztyn et al. (2025) and Sunstein (2026) supply the product trap framework used in the thesis to explain continued adoption despite unresolved trust.

Baek, C., Tate, T., & Warschauer, M. (2024). "ChatGPT seems too good to be true": College students' use and perceptions of generative AI. *Computers and Education: Artificial Intelligence*, 7, 100294. <https://doi.org/10.1016/j.caeai.2024.100294>

This quantitative study surveyed undergraduate students at a large U.S. research university to examine patterns of generative AI use and associated perceptions. Findings revealed that students frequently used ChatGPT for academic tasks while simultaneously expressing uncertainty about its reliability, a tension the authors frame as "too good to be true" skepticism. The study documents that even students who doubted ChatGPT's accuracy continued to use it, a behavioral paradox that is central to the trust-adoption gap explored in the thesis. This source contributes direct empirical evidence for the disconnect between skepticism and sustained use, supporting the thesis's argument that ChatGPT adoption is not contingent on resolved trust.

Bursztyn, L., Handel, B. R., Jiménez-Durán, R., & Roth, C. (2025). When product markets become collective traps: The case of social media. *American Economic Review*, 115(12), 4105–4136. <https://doi.org/10.1257/aer.20231468>

This economics study documents that many social media users would pay to have a platform removed if all peers were also removed, yet continue using it under current

conditions; a pattern the authors characterize as a "collective trap." The study provides empirical grounding for the theoretical concept of product traps, demonstrating that continued use can reflect external social dependencies rather than individual preference or satisfaction. In the thesis, this source is invoked alongside Sunstein (2026) to contextualize why undergraduate students continue adopting ChatGPT even when trust remains underdeveloped, suggesting that competitive academic pressure may sustain use independent of confidence in the tool.

Bushweller, K. (2024). Where does AI belong in education? Teachers and administrators have some strong opinions. *Education Week*, 43(21), 4–5.

This practitioner-oriented article surveys faculty and administrators about their attitudes toward generative AI in K–12 and higher education settings. The piece documents a prevailing institutional skepticism rooted in concerns about academic integrity, with many educators favoring prohibition over integration. For the thesis, this source illustrates the misalignment between institutional responses (largely restrictive) and student behavior (widespread adoption), contextualizing the problem statement and motivating the call for empirically informed AI literacy frameworks rather than reflexive restriction.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.

This foundational study introduced the Technology Acceptance Model (TAM), proposing that technology adoption is determined primarily by two perceptual variables: perceived usefulness (PU), the degree to which a user believes a technology will enhance performance, and perceived ease of use (PEOU), the degree to which they believe it will

require minimal effort. Davis developed and validated scales for both constructs through confirmatory factor analysis and field studies with IBM employees. TAM is the primary theoretical framework of the thesis, structuring the survey instrument, the qualitative themes, and the interpretive analysis. The thesis ultimately extends Davis's model by repositioning trust as a gating variable rather than a parallel construct.

Digital Education Council. (2024). *Global AI student survey 2024*.

<https://www.digitaleducationcouncil.com/>

This international survey report presents data on AI adoption patterns among higher education students across 16 countries, finding that 86% already use AI in their studies, with over half doing so weekly. Regional data indicate even higher rates in Latin America (92% adoption, 67% using AI daily or weekly for school). Critically, the report also finds that 58% of students globally report feeling they lack sufficient AI knowledge to use these tools effectively. For the thesis, this source provides the macro-level context for the study's problem statement, establishing that rapid AI adoption does not imply competency and justifying the need for empirical, behaviorally grounded AI literacy research.

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, K. A., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., ... Wright, R. (2023). So what if ChatGPT wrote it? Multidisciplinary perspectives on opportunities, challenges, and implications of generative conversational AI for research, practice, and policy. *International Journal of Information Management*, 71, 102642.

<https://doi.org/10.1016/j.ijinfomgt.2023.102642>

This multidisciplinary commentary article brings together perspectives from fields including information management, education, marketing, and public policy to assess the implications of generative AI tools, particularly ChatGPT, for professional and academic practice. The contributors collectively document that students are incorporating generative AI into academic workflows with or without institutional guidance, and that institutions have largely responded with apprehension rather than evidence-based frameworks. For the thesis, this source reinforces the problem statement by documenting the gap between student practice and institutional response on a global scale, supporting the call for empirical research on AI adoption.

Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, 27(1), 51–90. <https://doi.org/10.2307/30036519>

This highly influential study extended the Technology Acceptance Model by adding trust as a third construct alongside perceived usefulness and perceived ease of use, arguing that trust is especially critical in environments where users rely on system-generated outputs they cannot independently verify. Gefen and colleagues operationalized trust across multiple dimensions, including competence, benevolence, and integrity, and demonstrated its explanatory power in e-commerce adoption. For the thesis, this source is one of two foundational theoretical pillars (alongside Davis, 1989), and the trust extension it introduces forms the basis for the thesis's central argument that trust functions as a gating variable rather than a coequal TAM construct.

Gillespie, N., Lockey, S., Ward, T., Macdade, A., & Hassed, G. (2025). *Trust, attitudes and use of artificial intelligence: A global study 2025*. The University of Melbourne and KPMG. <https://doi.org/10.26188/28822919>

This large-scale international survey examined public attitudes toward and trust in artificial intelligence tools, finding that approximately two-thirds of the global public use AI regularly, with trust levels varying significantly by region, use case, and demographic. The report documents persistent concerns about AI accuracy and transparency even among frequent users. For the thesis, this source provides cross-national evidence that AI adoption outpaces trust, situating the thesis's findings within a broader global pattern and supporting the claim that the trust deficit observed among undergraduate students reflects wider societal dynamics rather than population-specific limitations.

Granić, A., & Marangunić, N. (2019). Technology acceptance model in educational contexts: A systematic literature review. *British Journal of Educational Technology*, 50(5), 2572–2593. <https://doi.org/10.1111/bjet.12864>

This systematic literature review surveyed over three decades of TAM applications in educational technology research, synthesizing findings on the explanatory power of perceived usefulness and perceived ease of use across diverse learning environments, student populations, and technology types. The authors confirm that TAM demonstrates robust predictive validity in educational contexts and identify trust and social influence as the most frequently added constructs in extended TAM models. For the thesis, this source validates the choice of TAM as the theoretical framework and situates the trust extension (Gefen et al., 2003) within an established lineage of educational technology adoption research.

Lambert, J., & Stevens, M. (2023). ChatGPT and generative AI technology: A mixed bag of concerns and new opportunities. *Computers in the Schools*, 41(1), 1–25. <https://doi.org/10.1080/07380569.2023.2256710>

This commentary article examines the implications of ChatGPT and generative AI for K–12 and higher education settings, weighing academic integrity concerns against pedagogical opportunities. The authors argue that institutional responses focused exclusively on prohibition miss opportunities to integrate AI as a legitimate learning tool and to develop students’ critical evaluation capacities. For the thesis, this source contributes to the problem statement by documenting the polarized institutional discourse around generative AI, supporting the argument that apprehension-based policy responses have obscured the need for empirical research into how students actually engage with these tools.

Sidoti, O., Park, E., & Gottfried, J. (2025, January 15). *About a quarter of U.S. teens have used ChatGPT for schoolwork—double the share in 2023*. Pew Research Center.

<https://www.pewresearch.org/short-reads/2025/01/15/about-a-quarter-of-us-teens-have-used-chatgpt-for-schoolwork-double-the-share-in-2023/>

This Pew Research Center report documents that the proportion of U.S. teenagers using ChatGPT for schoolwork doubled from 13% to 26% between 2023 and 2024, reflecting one of the fastest documented adoption rates of a new educational technology in recent decades. The data are drawn from a nationally representative survey of U.S. teens. For the thesis, this source provides the primary U.S.-specific adoption statistic cited in the introduction, establishing the urgency of the study by quantifying the pace and scale of student AI adoption and contextualizing the need for empirical research into what drives and shapes that adoption.

Sunstein, C. R. (2026). Social media, wasting time, and product traps. In J. F. Helliwell, R.

Layard, J. D. Sachs, J.-E. De Neve, L. B. Aknin, & S. Wang (Eds.), *World Happiness Report 2026* (pp. 163–176). <https://doi.org/10.18724/whr-4fk1-g247>

In this chapter, Sunstein introduces the concept of "product traps" to describe situations in which individuals continue using products they believe may reduce their own well-being, driven by external dependencies (such as peer use, social norms, or competitive pressure) rather than individual preference. Drawing on social media as the primary case, Sunstein argues that sustained use under such conditions reflects structural features of markets and social dynamics rather than personal satisfaction. For the thesis, the product trap framework is invoked to interpret why undergraduate students continue adopting ChatGPT despite an incomplete trust foundation, with the thesis proposing that competitive academic pressure and accessible early utility sustain adoption independent of resolved confidence in the tool's accuracy.

Tindle, R., Pozzebon, K., Willis, R., & Moustafa, A. A. (2023). *Academic misconduct and generative artificial intelligence: University students' intentions, usage, and perceptions*. PsyArXiv. <https://doi.org/10.31234/osf.io/d7e8u>

This preprint study examined undergraduate students' intentions to use generative AI for academically dishonest purposes, their actual usage patterns, and their perceptions of AI-generated content in academic contexts. The authors found that academic integrity concerns were salient for many students, though actual misuse was not uniformly distributed. For the thesis, Tindle et al. are cited alongside other sources in the problem statement to represent the dominant scholarly orientation toward generative AI as an integrity threat, and to establish that this institutional framing has oriented responses

toward restriction at the expense of empirical inquiry into how students actually engage with AI for legitimate academic purposes.

Yusuf, A., Pervin, N., & Román-González, M. (2024). Generative AI and the future of higher education: A threat to academic integrity or reformation? Evidence from multicultural perspectives. *International Journal of Educational Technology in Higher Education*, 21, 21. <https://doi.org/10.1186/s41239-024-00453-6>

This mixed-methods study gathered perspectives from students and educators across multiple national and cultural contexts to assess whether generative AI constitutes a threat to academic integrity or an opportunity for educational transformation. Findings reflected significant cross-cultural variation in attitudes, with some contexts welcoming AI as a learning scaffold and others emphasizing misuse risks. For the thesis, Yusuf et al. are cited in the problem statement to underscore the global scope of the misalignment between student AI adoption and institutional policy response, reinforcing the urgency of the research.

Category 4: Trust and Perception in Generative AI

These sources examine how users form, calibrate, and act on trust judgments with generative AI. Gefen et al. (2003) provide the TAM trust extension; Baek and Kim (2023) established that AI trust is dynamic and experience-dependent rather than a fixed prior disposition; and Baek et al. (2024) documented the “too good to be true” skepticism pattern that is a direct empirical antecedent to the thesis’s findings. Bender et al. (2021) explain the technical mechanism behind AI hallucination, grounding participants’ reports of fabricated citations in the structural properties of LLMs. Mayerhofer et al. (2025) and Gillespie et al. (2025) situate these dynamics comparatively (the former through observed tool-switching behavior, the latter through

global survey data), confirming that the trust deficit documented in the thesis reflects broader patterns rather than sample-specific limitations.

Baek, C., Tate, T., & Warschauer, M. (2024). “ChatGPT seems too good to be true”: College students’ use and perceptions of generative AI. *Computers and Education: Artificial Intelligence*, 7, 100294. <https://doi.org/10.1016/j.caeai.2024.100294>

This quantitative study surveyed undergraduate students at a large U.S. research university to examine patterns of generative AI use and associated perceptions. Findings revealed that students frequently used ChatGPT for academic tasks while simultaneously expressing uncertainty about its reliability, a tension the authors frame as “too good to be true” skepticism. Even students who doubted ChatGPT’s accuracy continued to use it, a behavioral paradox that is central to the trust-adoption gap explored in the thesis. This source contributes direct empirical evidence for the disconnect between skepticism and sustained use, supporting the thesis’s argument that ChatGPT adoption is not contingent on resolved trust.

Baek, T. H., & Kim, M. (2023). Is ChatGPT scary good? How user motivations affect creepiness and trust in generative artificial intelligence. *Telematics and Informatics*, 83, 102030. <https://doi.org/10.1016/j.tele.2023.102030>

This study investigated how user motivations, including the desire for task efficiency and personalized responses, shape perceptions of trust and “creepiness” toward ChatGPT.

The authors found that users were more likely to trust the tool when it demonstrated competence and adapted to individual needs, suggesting that trust is cultivated through satisfying interactions rather than being a stable prior disposition. For the thesis, this source supports the argument that trust in AI is dynamic and experience-dependent, and informs the discussion of how conversational, synthesized responses may build or erode user confidence across different task contexts.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623).

<https://doi.org/10.1145/3442188.3445922>

This foundational and frequently cited conference paper critiques large language models (LLMs), arguing that scale alone does not confer understanding or accuracy, and that LLMs generate fluent, plausible-sounding outputs that may nonetheless be factually incorrect or socially harmful. The authors introduce the term “stochastic parrot” to describe systems that produce statistically probable text without grounding in meaning or truth. For the thesis, this source provides the theoretical basis for understanding AI hallucination and fabricated citations, supporting the discussion of why ChatGPT produces confident-sounding outputs that students (and educators) cannot readily verify.

Gefen, D., Karahanna, E., & Straub, D. W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, 27(1), 51–90. <https://doi.org/10.2307/30036519>

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as a third construct alongside perceived usefulness and perceived ease of use, arguing that trust is especially critical in environments where users rely on system-generated outputs they cannot independently verify. Gefen and colleagues operationalized trust across multiple dimensions, including competence, benevolence, and integrity, and demonstrated its explanatory power in e-commerce adoption. For the thesis, this source is one of two foundational theoretical pillars (alongside Davis, 1989), and the trust extension it introduces forms the basis for the thesis's central argument that trust functions as a gating variable rather than a coequal TAM construct.

Gillespie, N., Lockey, S., Ward, T., Macdade, A., & Hassed, G. (2025). *Trust, attitudes and use of artificial intelligence: A global study 2025*. The University of Melbourne and KPMG. <https://doi.org/10.26188/28822919>

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Mayerhofer, K., Capra, R., & Elsweller, D. (2025). *Blending queries and conversations: Understanding tactics, trust, verification, and system choice in web search and chat interactions*. arXiv. <https://arxiv.org/abs/2504.05156>

This empirical study compared user behaviors during web search and chat-based AI

interactions, finding that users often accepted AI-generated responses without verification when outputs appeared well-formatted and plausible, even when they acknowledged uncertainty about accuracy. The study also documented patterns of tool-switching, with users returning to traditional search to verify AI outputs. For the thesis, this source provides direct empirical support for two central findings: that source transparency is the primary mechanism through which trust is regulated, and that ChatGPT's fluent synthesis can paradoxically undermine informed trust by presenting outputs with no visible verification pathway.

Category 5: Research Design and Methodology

These methodological texts authorized each major design decision in the study. Creswell (2008) and Kimmons (n.d.) justify the embedded mixed-methods design; Merriam and Tisdell (2016) established qualitative inquiry as appropriate for capturing the subjective and behavioral dimensions central to the research questions; and Ericsson and Simon (1993) authorized concurrent think-aloud protocols as valid cognitive data. Marshall (1996) and Hennink and Kaiser (2022) together justified the purposeful sampling strategy and ten-participant sample on the grounds of qualitative depth and thematic saturation. Braun and Clarke (2006), Saldaña (2021), and Glaser and Strauss (1967) provided the frameworks for thematic analysis, qualitative coding, and the constant comparative method used across the three analytical phases. Lincoln and Guba (1985) supplied the trustworthiness criteria (including member-checking and peer debriefing), and Garland (1991) provided the psychometric basis for the six-point Likert scale. Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>

This methodological text establishes thematic analysis as a rigorous qualitative research

method, distinguishing it from other forms of content analysis and providing a step-by-step framework for generating, reviewing, and naming themes across qualitative data. The authors emphasize transparency, reflexivity, and iterative coding as hallmarks of credible thematic work. In the thesis, Braun and Clarke's framework serves as the primary analytical method for interpreting participants' verbal explanations, interview responses, and think-aloud protocols. It underpins the generation of the eight themes reported in the Results and Discussion section.

Creswell, J. W. (2008). *Educational research: Planning, conducting, and evaluating quantitative and qualitative research* (3rd ed.). Pearson.

This widely used methods textbook provides comprehensive guidance on research design for educational studies, with detailed coverage of both quantitative and qualitative approaches and, critically for this thesis, mixed-methods designs. Creswell described embedded mixed-methods designs as those in which one method is dominant and a secondary method is nested within it to provide supplementary insight. The thesis draws explicitly on this framework to justify its embedded design structure, in which thematic analysis of qualitative data is the dominant strand and Likert-scale perception ratings serve an embedded, supportive role.

Ericsson, K. A., & Simon, H. A. (1993). *Protocol analysis: Verbal reports as data* (Rev. ed.). MIT Press.

This foundational methodology text establishes the theoretical and empirical basis for using verbal protocols (particularly concurrent think-aloud procedures) to collect valid data in cognitive research. The authors argue that verbalizations produced during task performance capture working memory contents in real time and are therefore a more

direct window into cognitive processes than post-hoc retrospective reports. For the thesis, Ericsson and Simon's framework justifies the use of concurrent think-aloud protocols during information-retrieval tasks, providing the methodological warrant for treating participants' verbalizations as evidence of their real-time trust evaluations and search decision-making.

Garland, R. (1991). The mid-point on a rating scale: Is it desirable? *Marketing Bulletin*, 2, 66–70.

This methodological article examines the psychometric and interpretive consequences of including or excluding a neutral midpoint on Likert-type rating scales. Garland argues that a neutral midpoint enables non-attitudes and socially desirable non-committal responses, reducing scale validity, and that forced-choice even-numbered scales better elicit directional preferences. For the thesis, this source provides the explicit justification for adopting a six-point rather than five-point Likert scale, ensuring that participants were required to indicate a directional position on each perception item rather than defaulting to a neutral response.

Glaser, B. G., & Strauss, A. L. (1967). *The discovery of grounded theory: Strategies for qualitative research*. Aldine.

This foundational qualitative methods text introduced grounded theory and the constant comparative method as systematic approaches to theory building from empirical data. The authors argued that theoretical insights should emerge inductively from the data through iterative comparison across cases, rather than being imposed deductively from prior frameworks. For the thesis, the constant comparative method is explicitly employed in the third phase of data analysis, used to refine open codes into overarching

observational themes by examining recurring patterns across all ten participants' session recordings and interview transcripts.

Hennink, M., & Kaiser, B. N. (2022). Sample sizes for saturation in qualitative research: A systematic review of empirical tests. *Social Science & Medicine*, 292, Article 114523. <https://doi.org/10.1016/j.socscimed.2021.114523>

This systematic review examined empirical studies that tested the relationship between sample size and thematic saturation in qualitative research, finding that meaningful saturation is typically achieved with 9–17 interviews when the sample is purposefully selected and relatively homogeneous. The authors provide guidance on determining appropriate sample sizes for qualitative inquiry based on research goals, analytical depth, and population specificity. For the thesis, this source justifies the use of a 10-participant sample by demonstrating that this size is sufficient to achieve saturation in qualitative research with a focused, purposefully recruited student population.

Kimmons, R. (n.d.). Mixed methods. In R. Kimmons, *Education research*. EdTech Books. https://edtechbooks.org/education_research/mixed_methods

This open-access chapter provides a practical overview of mixed-methods research designs in educational research, covering the logic, typology, and implementation of designs that integrate quantitative and qualitative methods. The chapter includes accessible descriptions of embedded, convergent, and explanatory designs. For the thesis, Kimmons is cited alongside Creswell (2008) to support the methodological rationale for the embedded mixed-methods design, in which a dominant qualitative strand (thematic analysis) is supported by an embedded quantitative strand (Likert-scale perception ratings) rather than treated as two equivalent research strands.

Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry*. Sage.

This seminal text established the criteria for rigor in qualitative research, proposing credibility, transferability, dependability, and confirmability as qualitative counterparts to the quantitative standards of internal validity, external validity, reliability, and objectivity. The authors introduced member-checking and peer debriefing as specific techniques for strengthening credibility. For the thesis, Lincoln and Guba are invoked in the data analysis section to justify the use of member-checking and peer debriefing as trustworthiness strategies, directly addressing the interpretive nature of thematic analysis and the potential for researcher bias.

Marshall, M. N. (1996). Sampling for qualitative research. *Family Practice*, 13(6), 522–526.

This methodological paper provides guidance on sampling strategies for qualitative health research, with particular attention to purposeful sampling as a method for identifying information-rich participants who can illuminate the phenomenon of interest in depth. Marshall distinguishes purposeful sampling from random or convenience sampling, emphasizing that qualitative inquiry seeks insight rather than representativeness. For the thesis, Marshall is cited as the methodological warrant for the purposeful sampling approach used to recruit participants with prior experience using both Google and ChatGPT, consistent with the qualitative emphasis of the embedded mixed-methods design.

Merriam, S. B., & Tisdell, E. J. (2016). *Qualitative research: A guide to design and implementation* (4th ed.). Jossey-Bass.

This comprehensive qualitative research methods text covers the design, execution, and interpretation of qualitative inquiry across multiple traditions, including case study,

grounded theory, and phenomenology. Merriam and Tisdell emphasize that qualitative methods are particularly well suited to capturing subjective experience, attitudinal nuance, and behavioral variability, dimensions that quantitative approaches alone cannot address. For the thesis, this source is cited in both the purpose statement and the methods section to justify the prioritization of qualitative methods and to establish that thematic analysis is an appropriate and rigorous approach for understanding perceptual and behavioral dimensions of AI tool use.

Saldaña, J. (2021). *The coding manual for qualitative researchers* (4th ed.). Sage.

This widely used methods manual provides detailed guidance on coding qualitative data, covering dozens of coding strategies suited to different research purposes, from descriptive and process codes to pattern and values coding. Saldaña emphasizes the importance of maintaining research memos and documenting initial analytic impressions to preserve interpretive proximity to the data. For the thesis, Saldaña is cited specifically to support the use of research memos and observational notes as the basis for initial coding in the third phase of data analysis, ensuring that emergent patterns were systematically documented before formal code application.

APPENDIX B

Consent/Institutional Review Board Approval Letter*Verbal Consent Script*

Hello, my name is Annie K. Brown, and I'm a researcher at Brigham Young University. Today, I'd like to invite you to participate in our research study titled 'Exploring Generative AI Perceptions and Information Retrieval Behaviors Among Students: An Investigation of Google Search and ChatGPT Queries.'

Today, we will ask you to share your experiences with traditional search engines and generative AI by completing common information-retrieval tasks and answering questions. We'll discuss how you typically find information and engage in information tasks using both Google and ChatGPT. Your participation is entirely voluntary, and you can withdraw at any time without consequences.

All the information you provide will be kept strictly confidential, and your identity will be protected. Your participation will contribute valuable insights to our research on students' information-seeking behaviors. Do you have any questions about the study or what participation entails?

Additionally, we would like to record this interview for research purposes. Your responses will remain confidential, and the recording will be used solely for analysis and reporting. May we have your permission to record this interview?

If at any point during the study or after the interview you have questions, feel free to contact me at [researcher's email]. For concerns about your rights as a research participant, you can contact the BYU Human Research Protection Program at 801-422-1461 or email BYU.HRPP@byu.edu.

IRB Approval Letter



Memorandum

To: Leanna Fry
 Department: BYU - HBLL - Harold B. Lee Library
 From: Sandee Aina, MPA, HRPP Associate Director
 Wayne Larsen, MAcc, IRB Administrator
 Date: October 02, 2024
 IRB#: IRB2024-266
 Title: Exploring GenAI Perceptions & Information Seeking Behaviors Among Students: A Comparative Analysis of Google and ChatGPT

Brigham Young University's IRB has approved the research study referenced in the subject heading as exempt, Categories 2 and 3. This study does not require an annual continuing review. Each year, near the anniversary of the approval date, you will receive an email reminding you of your obligations as an investigator and to check on the status of the study. You will receive this email each year until you close the study.

The study is approved as of 10/02/2024. Please reference your assigned IRB identification number in any correspondence with the IRB.

Continued approval is conditional upon your compliance with the following requirements:

1. A copy of the approved informed consent statement can be found in iRIS. No other consent statement should be used. Each research subject must be provided with a copy or a way to access the consent statement.
2. Any modifications to the approved protocol must be submitted, reviewed, and approved by the IRB before modifications are incorporated into the study.
3. All recruiting tools must be submitted and approved by the IRB before use.
4. Instructions to access approved documents, submit modifications, and report adverse events can be found on the IRB website, iRIS guide: <https://irb.byu.edu/iris-training-resources>
5. All non-serious unanticipated problems should be reported to the IRB within two weeks of the PI's first awareness of the problem. Prompt reporting is important, as unanticipated problems often require some modification of study procedures, protocols, and/or informed consent processes. Such modifications require the review and approval of the IRB. Please refer to the [IRB website](#) for more information.